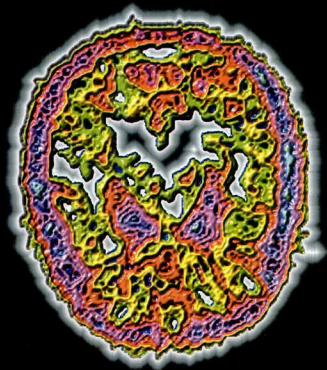


6th Edition



A Short Textbook of **PSYCHIATRY**

Niraj Ahuja

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Diagnosis and Classification in Psychiatry

Classification is a process by which the complex phenomena are organized into categories, classes or ranks so as to bring together those things that most resemble each other, and to separate those that differ.

Any classification of psychiatric disorders, like that of medical illnesses, should ideally be based on etiology. For a large majority of psychiatric disorders, no distinct etiology is known at present, although there are many attractive probabilities for each of them.

Hence, the only rational way to classify at present is *syndromal*. A *syndrome* is defined as a group of symptoms and signs that often occur together, and delineate a recognizable clinical condition.

This approach of classifying psychiatric disorders on the basis of their clinical signs and symptoms is very similar to the historical approach of classifying medical illnesses, when the etiology of a majority of medical illnesses was obscure.

There are three major purposes of classification:

1. To enable *communication* regarding the diagnosis.
2. To facilitate *comprehension* of the underlying causes.
3. To aid *prediction* of the prognosis.

This syndromal approach of classification, in the absence of known etiologies, fulfils these purposes reasonably well.

Before proceeding to look at the classification of psychiatric disorders, it is important to define what is a psychiatric disorder.

DEFINITION OF A PSYCHIATRIC DISORDER

The simplest way to conceptualize a psychiatric disorder is as a disturbance of

- i. Cognition (i.e. Thought), or
- ii. Conation (i.e. Action), or
- iii. Affect (i.e. Feeling), or

any disequilibrium between the three domains.

Another way to define a *psychiatric disorder* is as a clinically significant psychological or behavioral syndrome that causes significant distress (subjective symptomatology), disability (objective symptomatology), or loss of freedom; and which is not merely a socially deviant behavior or an expected response to a stressful life event (e.g. loss of a loved one).

Conflicts between the society and the individual are not considered mental disorders. A mental disorder should be a manifestation of behavioral, psychological, and/or biological dysfunction in that person (Definition modified after DSM-IV-TR).

Although slightly lengthy, this definition defines a psychiatric disorder more accurately.

NORMAL MENTAL HEALTH

According to World Health Organization (WHO), *health* is a state of complete physical, mental and social wellbeing, and not merely absence of disease or infirmity.

Normal mental health, much like normal health, is a rather difficult concept to define. There are several models for understanding what constitutes 'normality'.

1. **Medical Model (Normality as Health)** Normal mental health is conceptualized as the absence of psychiatric disorder ('disease') or psychopathology.
2. **Statistical Model (Normality as an Average)** Statistically normal mental health falls within two standard deviations (SDs) of the normal distribution curve.
3. **Utopian Model (Normality as Utopia)** In this model, the focus in defining normality is on 'optimal functioning'.
4. **Subjective Model:** According to this model, normality is viewed as an absence of distress, disability, or help-seeking behavior resulting thereof. This definition is similar in many ways to the medical model.
5. **Social Model:** A normal person, according to this definition, is expected to behave in a socially 'acceptable' behavior.
6. **Process Model (Normality as a Process)** This model views normality as a dynamic and changing process, rather than as a static concept. This model can be combined with any other model mentioned here.
7. **Continuum Model (Normality as a Continuum)** Normality and mental disorder are considered by this model as falling at the two ends of a continuum, rather than being disparate entities. According to this model, it is the severity (scores above the 'cut-off') that

determines whether a particular person's experience constitutes a symptom of a disorder or falls on the healthy side of the continuum.

Although, normality is not an easy concept to define, some of the following traits are more commonly found in 'normal' individuals.

1. Reality orientation.
2. Self-awareness and self-knowledge.
3. Self-esteem and self-acceptance.
4. Ability to exercise voluntary control over their behavior.
5. Ability to form affectionate relationships.
6. Pursuance of productive and goal-directive activities.

CLASSIFICATION IN PSYCHIATRY

Like any growing branch of medicine, Psychiatry has seen rapid changes in classification to keep up with a conglomeration of growing research data dealing with epidemiology, symptomatology, prognostic factors, treatment methods and new theories for the causation of psychiatric disorders.

Although the first attempts to classify mental illnesses can be traced back to Ayurveda, Plato (4th century BC) and Asclepiades (1st century BC), Classification in Psychiatry has certainly evolved ever since.

At present, there are two major classifications in Psychiatry, namely ICD-10 (1992) and DSM-IV-TR (2000).

1. **ICD-10** (International Classification of Diseases, 10th Revision, 1992) is WHO's classification for all diseases and related health problems (and not only psychiatric disorders).

TABLE 1.1. SOME VERSIONS OF ICD-10

1. Clinical Descriptions and Diagnostic Guidelines (CDDG)
2. Diagnostic Criteria for Research (DCR)
3. Multi-axial Classification Version
4. Primary Care Version

bances of thought, perception, affect, and/or behavior.

The disorders in this section include schizophrenia, schizotypal disorder, persistent delusional disorders, acute and transient psychotic disorders, induced delusional disorder, and schizo-affective disorders.

F31-F39 Mood (Affective) Disorders

This group includes mental and behavioral disorders characterized by a prominent disturbance of mood.

The disorders in this section include manic episode, depressive episode, bipolar affective disorder, recurrent depressive disorder, and persistent mood disorder.

F40-F48 Neurotic, Stress-related and Somatoform Disorders

(There is no category with code number F49)

This group includes mental and behavioral disorders that were earlier labeled as neurotic disorders, with an emphasis on psychological causation.

The disorders in this section include anxiety disorders, phobic anxiety disorders, obsessive-compulsive disorder, dissociative (conversion) disorders, somatoform disorders, reaction to stress, and adjustment disorders, and other neurotic disorders.

F50-F59 Behavioral Syndromes Associated with Physiological Disturbances and Physical Factors

This group includes mental and behavioral disorders that were earlier called psychosomatic disorders. The term psychosomatic is no longer used because its use implied that the association between the psychological factors and physical disorder is etiological. It also presumed that the psychological factors are not important in other medical illnesses and physical disorders.

The disorders in this section include eating disorders, non-organic sleep disorders, sexual dysfunctions (not caused by organic disorder or disease), mental and behavioral disorders associated with puerperium, and abuse of non-dependence-producing substances.

F60-F69 Disorders of Adult Personality and Behavior

This group includes mental and behavioral disorders that are the persistent expression of an individual's characteristic lifestyle and mode of relating to self and others.

The disorders in this section include specific personality disorders, enduring personality changes, habit and impulse disorders, gender-identity disorders, disorders of sexual preference, and psychological and behavioral disorders associated with sexual development and orientation.

F70-F79 Mental Retardation

This group includes disorders with arrested or incomplete development of the intellectual abilities and adaptive behavior, which may or may not be associated with other physical or mental disorder.

The disorders in this section include mild, moderate, severe, and profound mental retardation.

F80-F89 Disorders of Psychological Development

This group includes mental and behavioral disorders with an onset during infancy or childhood and characterized by an impairment or delay in the development of functions that are strongly related to biological maturation of the central nervous system.

The disorders in this section include specific developmental disorders of speech and language, specific developmental disorders of

scholastic skills, specific developmental disorders of motor function, mixed specific developmental disorders, and pervasive developmental disorders.

F90-F98 Behavioral and Emotional Disorders with Onset Usually Occurring in Childhood and Adolescence

This group includes miscellaneous mental and behavioral disorders that have an onset in childhood and adolescence.

The disorders in this section include hyperkinetic disorders, conduct disorders, mixed disorders of conduct and emotions, tic disorders, and other disorders.

F99 Unspecified Mental Disorder

MULTI-AXIAL CLASSIFICATION

The process of making a diagnosis is a useful

clinical exercise as the management is dependent on making a correct diagnosis. However, sometimes making a clinical diagnosis can lead to labeling the patient and can be stigmatizing. This degrades the individual patient to "just another case" and does not direct attention to the whole individual.

In the last three decades, there has been an upsurge of interest in multi-axial systems for achieving a more comprehensive description of an individual's clinical problems. The pattern adopted by DSM-IV-TR is a very good example. In this system, an individual patient is diagnosed on five separate axes (Table 1.2).

This method helps in making a more holistic, biopsychosocial assessment of an individual patient. Recently, ICD-10 has also brought out its own multi-axial classification version (Table 1.1).

2

Psychiatric History and Examination

Familiarity with the technique of *psychiatric assessment* is important not only for a psychiatrist but also for a medical specialist or practitioner, since more than one-third of the medical patients have psychiatric symptoms.

INTERVIEW TECHNIQUE

In no other branch of Medicine is the history taking interview as important as in Psychiatry. Most physicians have to communicate with their patients and they are never really taught how to talk with them. A skillful interview can clearly help in obtaining better information, making a more accurate diagnosis, establishing a better rapport with patients, and ensuring a greater treatment adherence.

A psychiatric interview is different from the routine medical interview in several ways (Table 2.1). A few important points regarding the interview technique are mentioned below. These serve as pointers towards a technique which clearly has to be mastered over a period of time with repeated examinations.

A *consistent scheme* should be used each time for recording the interview, although the interview need not (and should not) follow a fixed and rigid method. The interview technique should have flexibility, varying according to the varying clinical circumstances.

Whenever possible, the patient should be seen first. When the account of historical information given by the patient and the informant(s) is different, it is useful to *record them separately*.

During the interview, the patient should always be put at ease and a warm, *empathic relationship* should be established.

In psychiatric assessment, history taking interview and mental status examination need not be always conducted separately (though, they must be recorded separately). During the assessment, the interviewer should observe the abnormalities in verbal and non-verbal communication and make a note of them.

It is helpful to record the patient's responses *verbatim*, rather than only naming them (e.g. rather than writing delusion of persecution, it is better to record verbatim: "my neighbour is trying to poison me"). It is best done in the patient's own spoken language, whenever possible.

It is useful to ask *open-ended* and *non-directive questions* (e.g. "how are you feeling today?") rather than asking direct, leading questions (e.g. "are you feeling sad at present?").

Arguably the most important interviewing skills are *listening*, and showing that you are interested in listening and attending to the patient. It is also important to remember that listening is an active, and not a passive, process.

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A Short Textbook of Psychiatry

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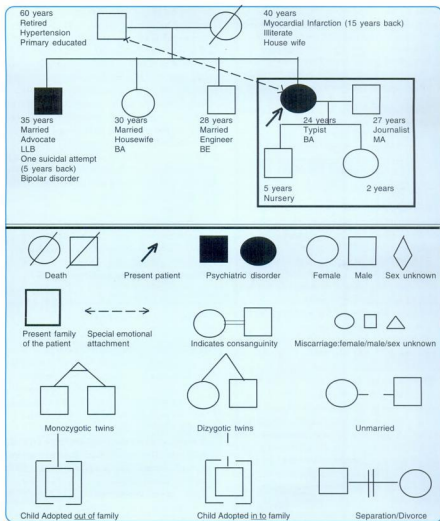


FIG. 2.1. A TYPICAL FAMILY TREE AND COMMON PEDIGREE SYMBOLS

1. *Family structure*: Drawing of a 'family tree' (pedigree chart) helps in recording all the relevant information in very little space and is easily readable. An example of a typical family tree is given in Fig. 2.1.

It should be noted whether the family is nuclear or extended nuclear (or joint) family. If consanguineous relationship is present, it should be recorded. The age and cause of death (if any) of the family members should be asked.

2. *A Family history* of similar or other psychiatric illnesses, major medical illnesses, alcohol or drug dependence and suicide (and suicidal attempts) should be recorded.

3. *Current social situation*: Home circumstances, per-capita income, socio-economic status, leader of the family (nominal as well as functional) and current attitude of the family members towards the patient's illness should be noted.

The communication patterns in the family, range of affectivity, cultural and religious values, and social support system should be enquired about, where relevant.

PERSONAL HISTORY

In a younger patient, it is often possible to give more attention to details regarding earlier personal history. In older patients, it is sometimes harder to get a detailed account of the early childhood history. Parents, if alive, can often provide much additional information regarding the past personal history.

The personal history can be recorded under the following headings:

1. Perinatal History

Any febrile illness; medications, drugs and/or alcohol use; trauma to abdomen and any physical or psychiatric illness during pregnancy (particularly in the first three months of

gestation) should be asked. Other relevant questions are: whether the patient was a wanted or unwanted child; date of birth; whether normal or abnormal delivery; any instrumentation; where born (hospital or home); any perinatal complications (cyanosis, convulsions, jaundice); APGAR score (if available); birth cry (immediate or delayed); any birth defects; any prematurity.

2. Childhood History

Whether the patient was brought up by mother or someone else; breast-feeding; weaning; any history suggestive of maternal deprivation, should be asked.

The age of passing each important developmental milestone should be noted. The age and ease of toilet training should be asked.

The occurrence of *neurotic traits* should be noted. These include stuttering, stammering, tics, enuresis, encopresis, night terrors, thumb sucking, nail biting, head banging, body rocking, morbid fears or phobias, somnambulism, temper tantrums, and food fads.

3. Educational History

The age of beginning and finishing formal education, academic achievements, and relationships with peers and teachers, should be asked.

Any school phobia, non-attendance, truancy, any learning difficulties and reasons for termination of studies (if occurs prematurely) should be noted.

4. Play History

The questions to be asked are, what games were played at what stage, with whom and where. Relationships with peers, particularly the opposite sex, should be recorded. The evaluation of play history is obviously more important in the younger patients.

5. Puberty

The age at menarche, and reaction to menarche (in females), the age at appearance of secondary sexual characteristics (in both females and males), nocturnal emissions (in males), masturbation and any anxiety related to changes in puberty should be asked.

6. Menstrual and Obstetric History

The regularity and duration of menses, the length of each cycle, any abnormalities, the last menstrual period, the number of children born, and termination of pregnancy (if any) should be asked for.

7. Occupational History

The age at starting work; jobs held in chronological order; reasons for changes; job satisfactions; ambitions; relationships with authorities, peers and subordinates; present income; and whether the job is appropriate to the educational and family background, should be asked.

8. Sexual and Marital History

Sexual information, how acquired and of what kind; masturbation (fantasy and activity); sex play, if any; adolescent sexual activity; premarital and extramarital sexual relationships, if any; sexual practices (normal and abnormal); and any gender identity disorder, are the areas to be enquired about.

The duration of marriage; time known the partner before marriage; marriage arranged by parents with or without consent, or by self-choice with or without parental consent; number of marriages, divorces or separations; role in marriage; interpersonal and sexual relations; contraceptive measures used; sexual satisfaction; mode and frequency of sexual intercourse; and psychosexual dysfunction (if any) should be asked.

Although conventionally the details of the 'family of procreation' are recorded here, they can also be recorded in the family tree.

9. Premorbid Personality (PMP)

It is important to elicit details regarding the personality of the individual (*temperament*, if the age is less than 16 years). Instead of using labels like schizoid or histrionic, it is more useful to describe the personality in some detail. The following subheadings are often used for the description.

i. Interpersonal relationship

Interpersonal relationships with family members, friends, and work colleagues; introverted / extraverted; ease of making and maintaining social relationships.

ii. Use of leisure time

Hobbies; interests; intellectual activities; critical faculty; energetic / sedentary.

iii. Predominant mood

Optimistic / pessimistic; stable / prone to anxiety; cheerful / despondent; reaction to stressful life events.

iv. Attitude to self and others

Self-confidence level; self-criticism; self-consciousness; self-centered / thoughtful of others; self-appraisal of abilities, achievements and failures.

v. Attitude to work and responsibility

Decision making; acceptance of responsibility; flexibility; perseverance; foresight.

vi. Religious beliefs and moral attitudes

Religious beliefs; tolerance of others' standards and beliefs; conscience; altruism.

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TABLE 2.2. MENTAL STATUS EXAMINATION

1. General Appearance and Behavior

- i. General Appearance
- ii. Attitude towards Examiner
- iii. Comprehension
- iv. Gait and Posture
- v. Motor Activity
- vi. Social Manner
- vii. Rapport

2. Speech

- i. Rate and Quantity
- ii. Volume and Tone
- iii. Flow and Rhythm

3. Mood and Affect**4. Thought**

- i. Stream and Form
- ii. Content

5. Perception**6. Cognition (Higher Mental Functions)**

- i. Consciousness
- ii. Orientation
- iii. Attention
- iv. Concentration
- v. Memory
- vi. Intelligence
- vii. Abstract thinking

7. Insight**8. Judgement****vii. Fantasy life**

Sexual and nonsexual fantasies; daydreaming-frequency and content; recurrent or favorite daydreams; dreams.

viii. Habits

Food fads; alcohol; tobacco; drugs; sleep.

One of the most reliable methods of assessment of premorbid personality is interviewing an informant familiar with the patient prior to the onset of illness.

PHYSICAL EXAMINATION

A detailed general physical examination (GPE) and systemic examination is a must in every

patient. Physical disease, which is etiologically important (for causing the psychiatric symptomatology), or accidentally co-existent, or secondarily caused by the psychiatric condition, is often present and can be detected by a good physical examination.

MENTAL STATUS EXAMINATION (MSE)*

Mental status examination is a standardized format in which the clinician records the psychiatric signs and symptoms present at the time of the interview.

MSE should describe all areas of mental functioning (Table 2.2). Some areas, however, may deserve more emphasis according to the clinical impressions that may arise from the history, e.g. mood and affect in depression, and cognitive functions in delirium and dementia.

** The definitions of some MSE terms are given in Appendix III.*

1. General Appearance and Behavior

A rich deal of information can be elicited from the examination of the general appearance and behavior. While examining, it is important to remember the sociocultural background and personality of the patient.

Understandably, the general appearance and behavior needs to be given more emphasis in the examination of an uncooperative patient.

i. General Appearance

The important points to be noted are:

- Physique and body habitus (build) and physical appearance (approximate height, weight, and appearance),
- Looks comfortable/uncomfortable,
- Physical health,
- Grooming, Hygiene, Self care,
- Dressing (adequate, appropriate, any peculiarities),
- Facies (non-verbal expression of mood),
- Effeminate/masculine.

The *affect* is similarly described under *quality of affect*, *range of affect* (of emotional changes displayed over time), *depth or intensity of affect* (normal, increased or blunted) and *appropriateness of affect* (in relation to thought and surrounding environment).

Mood is described as general warmth, euphoria, elation, exaltation and/or ecstasy (seen in severe mania) in *mania*; anxious and restless in *anxiety* and *depression*; sad, irritable, angry and/or despaired in *depression*; and shallow, blunted, indifferent, restricted, inappropriate and/or labile in *schizophrenia*.

Anhedonia may occur in both *schizophrenia* and *depression*.

4. Thought

Normal thinking is a goal directed flow of ideas, symbols and associations initiated by a problem or a task, characterized by rational connections between successive ideas or thoughts, and leading towards a reality oriented conclusion. Therefore, thought process that is not goal-directed, or not logical, or does not lead to a realistic solution to the problem at hand, is not normal.

Traditionally, in the clinical examination, thought is assessed (by the content of speech) under the four headings of *stream*, *form*, *content* and *possession* of thought. However, since there is widespread disagreement regarding this subdivision, 'thought' is discussed here under the following two headings of '*stream and form*', and '*content*'.

i. Stream and Form of Thought

For obvious reasons, the 'stream of thought' overlaps with examination of 'speech'. Spontaneity, productivity, flight of ideas, prolixity, poverty of content of speech, and thought block should be mentioned here.

The 'continuity' of thought is assessed; Whether the thought processes are relevant to the questions asked; Any loosening of associations,

tangentiality, circumstantiality, illogical thinking, perseveration, or verbigeration is noted.

ii. Content of Thought

Any preoccupations;

Obsessions (recurrent, irrational, intrusive, ego-dystonic, ego-alien ideas);

Contents of phobias (irrational fears);

Delusions (false, unshakable beliefs) or Over-valued ideas;

Explore for delusions/ideas of persecution, reference, grandeur, love, jealousy (infidelity), guilt, nihilism, poverty, somatic (hypochondriacal) symptoms, hopelessness, helplessness, worthlessness, and suicidal ideation.

Delusions of control, thought insertion, thought withdrawal, and thought broadcasting are Schneiderian first rank symptoms (SFRS). The presence of neologisms should be recorded here.

5. Perception

Perception is the process of being aware of a sensory experience and being able to recognize it by comparing it with previous experiences.

TABLE 2.3. CLINICAL RATING OF INSIGHT

Insight is rated on a 6-point scale from one to six.

1. Complete denial of illness.
2. Slight awareness of being sick and needing help, but denying it at the same time.
3. Awareness of being sick, but it is attributed to external or physical factors.
4. Awareness of being sick, due to something unknown in self.
5. *Intellectual Insight*: Awareness of being ill and that the symptoms/failures in social adjustment are due to own particular irrational feelings/thoughts; yet does not apply this knowledge to the current/future experiences.
6. *True Emotional Insight*: It is different from intellectual insight in that the awareness leads to significant basic changes in the future behavior and personality.

TABLE 2.5. DIAGNOSTIC FORMULATION

	Biological	Psychological	Social
Predisposing			
Precipitating			
Perpetuating			

and the time spent in hospital), *place* (test by asking the present location, building, city, and country) and *person* (test by asking his own name, and whether he can identify people around him and their role in that setting). Disorientation in time usually precedes disorientation in place and person.

iii. Attention

Is the attention easily aroused and sustained; Ask the patient to repeat digits forwards and backwards (*digit span test; digit forward and backward test*), one at a time (e.g. patient may be able to repeat 5 digits forward and 3 digits backwards). Start with two digit numbers, increasing gradually up to eight digit numbers or till failure occurs on three consecutive occasions.

iv. Concentration

Can the patient concentrate; Is he easily distractible; Ask to subtract serial sevens from hundred (*100-7 test*), or serial threes from fifty (*50-3 test*), or to count backwards from 20, or enumerate the names of the months (or days of the week) in the reverse order.

Note down the answers and the time taken to perform the tests.

v. Memory

a. Immediate Retention and Recall (IR and R)

Use the *digit span test* to assess the immediate memory; *digit forwards* and *digit backwards subtests* (also used for testing attention; are described under attention).



FIG. 2.3. ETIOLOGICAL FACTORS

b. Recent Memory

Ask how did the patient come to the room/hospital; what he ate for dinner the day before or for breakfast the same morning.

Give an address to be memorized and ask it to be recalled 15 minutes later or at the end of the interview.

c. Remote Memory

Ask for the date and place of marriage, name and birthdays of children, any other relevant questions from the person's past.

Note any amnesia (anterograde/retrograde), or confabulation, if present.

vi. Intelligence

Intelligence is the ability to think logically, act rationally, and deal effectively with the environment.

Ask questions about general information, keeping in mind the patient's educational and social background, his experiences and interests, e.g. ask about the current and the past prime ministers and presidents of India, the capital of India, and the name of the various states.

Test for reading and writing; Use simple tests of calculation.

vii. Abstract Thinking

Abstract thinking is characterized by the ability to:

- assume a mental set voluntarily,
- shift voluntarily from one aspect of a situation to another,

TABLE 3.1. DELIRIUM: IMPORTANT CAUSES

A. Metabolic causes

- i. Hypoxia, carbon dioxide narcosis.
- ii. Hypoglycemia.
- iii. Hepatic encephalopathy, uremic encephalopathy.
- iv. Cardiac failure, cardiac arrhythmias, cardiac arrest.
- v. Water and electrolyte imbalance (water, Na⁺, K⁺, Mg⁺⁺, Ca⁺⁺).
- vi. Metabolic acidosis or alkalosis.
- vii. Fever, anemia, hypovolemic shock.
- viii. Carcinoid syndrome, porphyria.

B. Endocrine causes

- i. Hypo- and hyper-pituitarism.
- ii. Hypo- and hyper-thyroidism.
- iii. Hypo- and hyper-parathyroidism.
- iv. Hypo- and hyper-adrenalism.

C. Drugs (both ingestion and withdrawal can cause delirium) and poisons

- i. Digitalis, quinidine, antihypertensives.
- ii. Alcohol, sedatives, hypnotics (especially barbiturates).
- iii. Tricyclic antidepressants, antipsychotics, anticholinergics, disulfiram.
- iv. Anticonvulsants, L-dopa, opiates.

- v. Salicylates, steroids, penicillin, insulin.
- vi. Methyl alcohol, heavy metals, biocides.

D. Nutritional deficiencies

- i. Thiamin, niacin, pyridoxine, folic acid, B₁₂.
- ii. Proteins.

E. Systemic infections

- i. Acute and chronic, e.g. septicemia, pneumonia, endocarditis.

F. Intracranial causes

- i. Epilepsy (including post-ictal states).
- ii. Head injury, subarachnoid hemorrhage, subdural hematoma.
- iii. Intracranial infections, e.g. meningitis, encephalitis, cerebral malaria.
- iv. Migraine.
- v. Stroke (acute phase), hypertensive encephalopathy.
- vi. Focal lesions, e.g. right parietal lesions (like abscess, neoplasm).

G. Miscellaneous

- i. Postoperative states (including ICU delirium).
- ii. Sleep deprivation.
- iii. Heat, electricity, radiation.

3. Psychomotor disturbances (hypo- or hyper-activity and unpredictable shifts from one to the other; increased reaction time; increased or decreased flow of speech; enhanced startle reaction);

TABLE 3.2. PREDISPOSING FACTORS IN DELIRIUM

1. Pre-existing brain damage or dementia.
2. Extremes of age (very old and very young).
3. Previous history of delirium.
4. Alcohol or drug dependence.
5. Generalized or focal cerebral lesion.
6. Chronic medical illness.
7. Surgical procedure and post-operative period.
8. Severe psychological symptoms, like fear.
9. Treatment with psychotropic medicines.
10. History of head injury.
11. Individual susceptibility to delirium.

4. Disturbance of sleep-wake cycle (insomnia or, in severe cases total sleep loss or reversal of the sleep-wake cycle; daytime drowsiness; nocturnal worsening of symptoms; disturbing dreams or nightmares, which may continue as hallucinations after awakening);
5. Emotional disturbances, e.g. depression, anxiety, fear, irritability, euphoria, apathy, or perplexity.

The onset is usually rapid, the course diurnally fluctuating, and the total duration of the condition much less than 6 months. The above clinical picture is so characteristic that a fairly confident diagnosis of delirium can be made even if the underlying cause is not clearly established.

TABLE 3.6. SOME COMMON CAUSES OF DEMENTIA

A. Parenchymatous brain disease

Alzheimer's disease, Pick's disease, Parkinson's disease, Huntington's chorea, Steel Richardson syndrome (progressive supranuclear palsy)

B. Vascular dementia

Multi-infarct dementia, sub-cortical vascular dementia (Binswanger's disease)

C. Toxic dementias

Bromide intoxication, drugs, heavy metals, alcohol, carbon monoxide, analgesics, anti-convulsants, benzodiazepines, psychotropic drugs

D. Metabolic dementias

Chronic hepatic or uremic encephalopathy, dialysis dementia, Wilson's disease

E. Endocrine causes

Thyroid, parathyroid, pituitary, adrenal dysfunction

F. Deficiency dementias

Pernicious anemia, pellagra, folic acid deficiency, thiamine deficiency

G. Dementias due to infections

Creutzfeldt-Jacob disease, neurosyphilis, chronic meningitis, viral encephalitis, AIDS dementia, other HIV-related disorders, sub-acute sclerosing panencephalitis

H. Neoplastic dementias

Neoplasms and other intracranial space-occupying lesions

I. Traumatic dementias

Chronic subdural hematoma, head injury

J. Hydrocephalic dementia

Normal pressure hydrocephalus

difficulty in sustaining attention and concentration, and reduced intellectual capacity.

In contrast, patients with dementia do not often complain of these disturbances. In fact, when confronted with evidence of memory impairment, they often confabulate.

As depression may often be superimposed on dementia, it is at times necessary to undertake a therapeutic trial with antidepressants, if the clinical picture is unclear.

It is useful to differentiate dementia in to cortical and subcortical subtypes (Table 3.5).

Etiology

A large number of conditions can cause dementia (Table 3.6). However, a majority of cases are due to a few common causes, like Alzheimer's disease, multi-infarct dementia, and hypothyroidism.

Five clinically important types of dementia are briefly discussed here.

1. Alzheimer's Dementia (Dementia of Alzheimer's Type; DAT)

This is the commonest cause of dementia, seen in about 70% of all cases of dementia in USA. It is more commonly seen in women. Earlier, it was differentiated into two forms: a presenile form and a senile form. Now it is known that these two forms represent the same disease clinically and pathologically. There is some evidence to suggest that Alzheimer's disease may have a genetic basis.

The diagnosis of DAT is by exclusion of all other causes of dementia, as there are no distinct diagnostic clinical features or laboratory investigations.

Autopsy shows enlarged cerebral ventricles, widened cerebral sulci and shrinkage of cerebral cortex (macroscopically); and senile plaques, neurofibrillary tangles, cortical nerve cell loss and granulovacuolar degeneration (microscopically). However, these changes are only quantitatively, and not qualitatively, different from normal aged brain. Neurochemically, there is a marked decrease in brain choline acetyltransferase (CAT) with a similar decrease in brain acetyl cholinesterase (AChE).

- v. Herpes simplex encephalitis.
- vi. Space occupying lesions in the region of III ventricle (e.g. neoplasms).

Management

1. Treatment of the underlying cause, e.g. thiamin (high doses) in Wernicke-Korsakoff syndrome. However, usually the treatment is of not much help, except in prevention of further deterioration and the prognosis is often poor.
2. Supportive care for general condition and the associated medical illness.

OTHER ORGANIC MENTAL DISORDERS (DUE TO BRAIN DAMAGE AND DYSFUNCTION, AND TO PHYSICAL DISEASES)

This group includes miscellaneous mental disorders which are causally related to brain dysfunction due to primary cerebral disease, systemic disease (secondary), or toxic substances.

According to the ICD-10 (CDDG), the following features are required for the diagnosis:

1. Evidence of cerebral disease, damage or dysfunction, or of systemic physical disease, known to be associated with one of the listed syndromes;
2. A temporal relationship (weeks or a few months) between the development of the underlying disease and the onset of the mental syndrome;
3. Recovery from the mental disorder following removal or improvement of the underlying presumed cause;
4. Absence of evidence to suggest an alternative cause of the mental syndrome (such as a strong family history or precipitating stress).

If conditions 1 and 2 are present, a provisional diagnosis can be made; if all four are present, the certainty of diagnosis is significantly increased.

The following clinical conditions are known to be associated with an increased risk for these other organic mental disorders:

1. Primary cerebral diseases

Epilepsy, limbic encephalitis, Huntington's disease, head trauma, brain neoplasms, vascular cerebral disease, cerebral malformations.

2. Systemic diseases

Extracranial neoplasms (e.g. carcinoma pancreas), collagen diseases (e.g. SLE), endocrine disease (e.g. hypothyroidism, hyperthyroidism, Cushing's disease), metabolic disorders (e.g. hypoglycemia, porphyria, hypoxia); infectious diseases (e.g. trypanosomiasis).

3. Drugs

Steroids, propranolol, levodopa, methylodopa, anti-hypertensives, antimalarials, alcohol and other psychoactive substances.

ORGANIC HALLUCINOSIS

According to the ICD-10 (CDDG), the following features are required for the diagnosis of organic hallucinosis, in addition to the general guidelines for the diagnosis of other organic mental disorders, described earlier:

1. Persistent or recurrent hallucinations due to an underlying organic cause. The hallucinations can occur in any sensory modality but are usually *visual* (most common) or auditory in nature. In many cases, they depend on the underlying cause. These *hallucinations* can range from very simple and unformed, to very complex and well-organized. Usually the patients realize that the hallucinations are not real but sometimes there may be a delusional elaboration of hallucinations.
2. No major disturbance of consciousness, intelligence, memory, mood or thought.

2. Altered emotional behavior, characterized by emotional lability, shallow and unwarranted cheerfulness (euphoria, inappropriate jocularity), and easy change to irritability or short-lived outbursts of anger and aggression; in some instances apathy may be a more prominent feature;
3. Expression of needs and impulses without consideration of the consequences or social convention (the patient may engage in dissocial acts, such as stealing, inappropriate sexual advances, or voracious eating, or may exhibit disregard for personal hygiene);
4. Cognitive disturbances, in the form of suspiciousness or paranoid ideation, and/or excessive preoccupation with a single, usually abstract, theme (e.g. religion, 'right and wrong');
5. Marked alteration of the rate and flow of language production, with features such as circumstantiality, over-inclusiveness, viscosity, and hypergraphia;
6. Altered sexual behavior (hyposexuality or change of sexual preference).

Etiology

1. *Temporal lobe epilepsy* (complex partial seizures) causing temporal lobe (personality) syndrome (see Table 3.4).
2. *Concussion* (post-concussional syndrome).
3. *Encephalitis* (post-encephalitis syndrome).
4. *Multiple sclerosis* (early).
5. *Cerebral neoplasms*, especially in frontal lobe (frontal lobe syndromes) and parietal lobe (see Table 3.4).
6. *Cerebrovascular disease*.
7. *Psychoactive drugs* (rarely).

TABLE 3.7. ORGANIC PERSONALITY DISORDERS

Organic Personality Disorder	Clinical Features
1. Frontal lobe syndrome (Types) a. Orbito-frontal syndrome (<i>Pseudo-psychopathic</i>) b. Frontal convexity type (<i>Pseudo-depressive</i>) c. Medial frontal syndrome (<i>Akinetic</i>)	Disinhibition, jocularity, impulsivity, impaired insight and judgement Apathy, lack of initiative, retardation, perseveration Akinesia, incontinence, poor verbal output
2. Temporal lobe syndrome	Egocentricity, explosive affect, perseveration, excessive religiosity, obsessional traits
3. Bilateral temporal lobe or limbic system lesions	Emotional placidity, hyper-orality, altered sexual behavior, excessive exploration of environment (<i>hyper-metamorphosis</i>).

Management

1. Treatment of the underlying cause, if treatable.
2. Symptomatic treatment, with lithium, carbamazepine for aggressive behavior and impulse dyscontrol, and/or antipsychotics (occasionally) for violent behavior may be needed.

MISCELLANEOUS ORGANIC MENTAL DISORDERS

Other organic mental disorders described in ICD-10 include organic dissociative disorder, organic emotionally labile (asthenic) disorder, and mild cognitive disorder.

TABLE 4.2. ETIOLOGICAL FACTORS IN SUBSTANCE USE DISORDERS**1. Biological Factors**

- i. Genetic vulnerability (family history of substance use disorder, e.g. in type II alcoholism)
- ii. Co-morbid psychiatric disorder or personality disorder
- iii. Co-morbid medical disorders
- iv. Reinforcing effects of drugs (explains continuation of drug use)
- v. Withdrawal effects and craving (explains continuation of drug use)
- vi. Biochemical factors (e.g. role of dopamine and norepinephrine in cocaine, ethanol and opioid dependence)

2. Psychological Factors

- i. Curiosity; need for novelty seeking
- ii. General rebelliousness and social non-conformity
- iii. Early initiation of alcohol and tobacco
- iv. Poor impulse control
- v. Sensation-seeking (high)
- vi. Low self-esteem (anomie)
- vii. Concerns regarding personal autonomy
- viii. Poor stress management skills
- ix. Childhood trauma or loss
- x. Relief from fatigue and/or boredom
- xi. Escape from reality
- xii. Lack of interest in conventional goals
- xiii. Psychological distress

3. Social Factors

- i. Peer pressure (often more important than parental factors)
- ii. Modeling (imitating behavior of important others)
- iii. Ease of availability of alcohol and drugs
- iv. Strictness of drug law enforcement
- v. Intrafamilial conflicts
- vi. Religious reasons
- vii. Poor social/familial support
- viii. 'Perceived distance' within the family
- ix. Permissive social attitudes
- x. Rapid urbanization.

Etiology

The various etiological factors in substance use disorders are briefly summarized in Table 4.2.

ALCOHOL USE DISORDERS

Alcohol dependence was previously called as *alcoholism*. This term, like 'addiction', has been dropped due to its derogatory meaning.

According to Jellinek, there are five 'species' of alcohol dependence ('*alcoholism*') on the basis of the patterns of use (and not on the basis of severity).

A. Alpha (α) Alcoholism

- i. Excessive and inappropriate drinking to relieve physical and/or emotional pain.
- ii. No loss of control.
- iii. Ability to abstain present.

B. Beta (β) Alcoholism

- i. Excessive and inappropriate drinking.

TABLE 4.3. CLASSIFICATION OF ALCOHOLISM

Factors	Type I	Type II
Synonym	Milieu-limited	Male-limited
Sex	Both sexes	Mostly in males
Age of onset	> 25 years	< 25 years
Etiological factors	Genetic factors important; strong environmental influences are contributory	Heritable; environmental influences are limited
Family history	May be positive	Parental alcoholism and antisocial behavior usually present
Loss of control	Present	No loss of control
Other features	Psychological dependence; and guilt present	Drinking followed by aggressive behavior; spontaneous alcohol seeking
Pre-morbid personality traits	Harm avoidance; high reward dependence	Novelty-seeking

alcohol use. Characteristically, the onset occurs after a period of persistent vomiting. The important clinical signs are:

- i. *Ocular signs:* Coarse nystagmus and ophthalmoplegia, with bilateral external rectus paralysis occur early. In addition, pupillary irregularities, retinal hemorrhages and papilledema can occur, causing an impairment of vision.
- ii. *Higher mental function disturbance:* Disorientation, confusion, recent memory disturbances, poor attention span and distractibility are quite common. Other early symptoms are apathy and ataxia. Peripheral neuropathy and serious malnutrition are often co-existent. Neuropathologically, neuronal degeneration and hemorrhage is seen in thalamus, hypothalamus, mammillary bodies and midbrain.

2. Korsakoff's psychosis

As Korsakoff's psychosis often follows Wernicke's encephalopathy, these are together referred to as the Wernicke-Korsakoff syndrome.

Clinically, Korsakoff's psychosis presents as an *organic amnestic syndrome*, characterized by gross memory disturbances with confabulation. Insight is often impaired.

The neuropathological lesion is usually widespread, but the most consistent changes are seen in bilateral dorsomedial nuclei of thalamus and mammillary bodies. The changes are also seen in periventricular and periaqueductal grey matter, cerebellum and parts of brain stem.

The cause is usually severe, untreated, thiamin deficiency secondary to chronic alcohol use.

3. Marchiafava-Bignami disease

This is a rare disorder characterized by disorientation, epilepsy, ataxia, dysarthria, hallucinations, spastic limb paralysis, and deterioration of personality and intellectual functioning. There

is a widespread demyelination of corpus callosum, optic tracts and cerebellar peduncles. The cause is probably an alcohol-related nutritional deficiency.

4. Other complications

These include:

- i. Alcoholic dementia.
- ii. Cerebellar degeneration.
- iii. Peripheral neuropathy.
- iv. Central pontine myelinolysis.

Treatment

Before starting any treatment, it is important to follow these steps:

- i. Ruling out (or diagnosing) any physical disorder.
- ii. Ruling out (or diagnosing) any psychiatric disorder.
- iii. Assessment of motivation for treatment.
- iv. Assessment of social support system.
- v. Assessment of personality characteristics of the patient.
- vi. Assessment of the current and past social, interpersonal and occupational functioning.

The treatment can be broadly divided into two categories, which are often interlinked. These are detoxification and treatment of alcohol dependence.

1. Detoxification

Detoxification is treatment of alcohol withdrawal symptoms, i.e. symptoms produced by the removal of the 'toxin' (alcohol). The best way to stop alcohol (or any other drug of dependence) is to stop it suddenly, unless the risks of acute discontinuation are felt to be very high by the treating team.

The usual duration of uncomplicated withdrawal syndrome is 7-14 days. The aim of detoxification is the symptomatic management of the emergent withdrawal symptoms.

TABLE 4.7. OPIOID DERIVATIVES

A. Natural Alkaloids of Opium

1. Morphine
2. Codeine
3. Thebaine
4. Noscapine
5. Papaverine

B. Synthetic Compounds

1. Heroin
2. Nalorphine
3. Hydromorphone
4. Methadone
5. Dextropropoxyphene
6. Meperidine (Pethidine)
7. Cyclazocine
8. Levallorphan
9. Diphenoxylate

The most important dependence producing derivatives are morphine and heroin. They both, like majority of dependence producing opioids, bind to μ (mu) opioid receptors. The other opioid receptors are κ (kappa, e.g. for pentazocine), δ (delta, e.g. for a type of enkephalin), σ (sigma e.g. for phencyclidine), ϵ (epsilon) and λ (lambda).

Heroin or diacetyl-morphine is about two times more potent than morphine in injectable form. Apart from the parenteral mode of administration, heroin can also be smoked or 'chased' (*chasing the dragon*), often in an impure form (called 'smack' or 'brown sugar' in India).

Heroin is more addicting than morphine and can cause dependence after a short period of exposure. Tolerance to heroin occurs rapidly and can be increased to up to more than 100 times the first dose needed to produce an effect.

Acute Intoxication

This is characterized by apathy, bradycardia, hypotension, respiratory depression, subnormal core body temperature and pin-point pupils. Later, delayed reflexes, thready pulse and coma

may occur in case of a large overdose. In severe intoxication, mydriasis may occur due to hypoxia.

Withdrawal Syndrome

The onset of withdrawal symptoms occurs typically within 12-24 hours, has a peak within 24-72 hours and symptoms subside within 7-10 days of the last dose of opioid.

The characteristic symptoms include lacrimation, rhinorrhea, pupillary dilation, sweating, diarrhea, yawning, tachycardia, mild hypertension, insomnia, raised body temperature, muscle cramps, generalized bodyache, severe anxiety, piloerection, nausea, vomiting and anorexia.

There are marked individual differences in the presentation of withdrawal symptoms. The heroin withdrawal syndrome is far more severe than the withdrawal syndrome seen with morphine.

Complications

The important complications of chronic opioid use may include one or more of the following:

1. Complications due to illicit drug (contaminants): Parkinsonism, degeneration of globus pallidus, peripheral neuropathy, amblyopia, transverse myelitis.
2. Complications due to intravenous use: AIDS, skin infection(s), thrombophlebitis, pulmonary embolism, septicemia, viral hepatitis, tetanus, endocarditis.
3. Drug peddling and involvement in criminal activities (social complication).

Treatment

Before treatment, a correct diagnosis must be made on the basis of history, examination (pin-point pupils during intoxication or withdrawal symptoms) and laboratory tests. These tests are:

1. *Naloxone challenge test* (to precipitate the withdrawal symptoms).

anovulatory cycles, reversible inhibition of spermatogenesis, blockade of gonadotropin releasing hormone, and increased risk for the developing fetus (if taken during pregnancy).

Treatment

As the withdrawal syndrome is usually very mild, the management consists of supportive and symptomatic treatment, if the patients comes to medical attention. The psychiatric symptoms may require appropriate psychotropic medication and sometimes hospitalization. Psychotherapy and psychoeducation are very important in the management of psychic dependence.

COCAINE USE DISORDER

Cocaine is an alkaloid derived from the coca bush, *Erythroxylum coca*, found in Bolivia and Peru. It was isolated by Albert Neimann in 1860 and was used by Karl Koller (a friend of Freud) in 1884 as the first effective local anesthetic agent.

Cocaine (common street name: *Crack*) can be administered orally, intranasally, by smoking (*free basing*) or parenterally, depending on the preparation available (Fig. 4.2). Cocaine HCl is the commonest form used, followed by the free base alkaloid. Both intravenous use and free base inhalation produce a 'rush' of pleasurable sensations.

Cocaine is a central stimulant which inhibits the reuptake of dopamine, along with the reuptake of norepinephrine and serotonin. In animals, cocaine is the most powerful reinforcer of the drug-taking behavior. A typical pattern of cocaine use is cocaine 'runs' (binges), followed by the cocaine 'crashes' (interruption of use). Cocaine is sometimes used in combination with opiates like heroin ('speed ball') or at times amphetamines. Previously uncommon, cocaine misuse appears to be recently a growing problem in the metros of India.

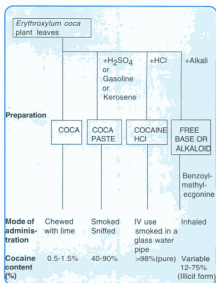


FIG. 4.2. COCAINE PREPARATIONS

Acute Intoxication

Acute cocaine intoxication is characterized by pupillary dilatation, tachycardia, hypertension, sweating, and nausea or vomiting. A hypomanic picture with increased psychomotor activity, grandiosity, elation of mood, hypervigilance and increased speech output may be present. Later, judgement is impaired and there is impairment of social or occupational functioning.

Withdrawal Syndrome

Cocaine use produces a very mild physical, but a very strong psychological, dependence. A triphasic withdrawal syndrome usually follows an abrupt discontinuation of chronic cocaine use (Table 4.9).

Complications

The complications of chronic cocaine use include acute anxiety reaction, uncontrolled

pentobarbital and amobarbital. Their use has recently decreased markedly as benzodiazepines have replaced barbiturates in the majority of their clinical uses.

Barbiturates produce marked physical and psychological dependence. Tolerance (both central and metabolic) develops rapidly and is usually marked. There is also a cross tolerance with alcohol.

Intoxication and Complications

Acute intoxication, typically occurring as an episodic phenomenon, is characterized by irritability, increased productivity of speech, lability of mood, disinhibited behavior, slurring of speech, incoordination, attentional and memory impairment, and ataxia. Mild barbiturate intoxication resembles alcohol intoxication; severe forms may present with diplopia, nystagmus, hypotonia, positive Romberg's sign and suicidal ideation. Drug automatism may sometimes lead to lethal accidents.

Intravenous use can lead to skin abscesses, cellulitis, infections, embolism and hypersensitivity reactions.

Withdrawal Syndrome

The barbiturate withdrawal syndrome can be very severe. It usually occurs in individuals who are taking more than 600-800 mg/day of secobarbital equivalent for more than one month.

It is usually characterized by marked restlessness, tremors, hypertension, seizures and, in severe cases, a psychosis resembling delirium tremens. The withdrawal syndrome is at its worst about 72 hours after the last dose. Coma, followed by death, can occur in some cases.

Treatment

The barbiturate intoxication should be treated symptomatically. If the patient is conscious, induction of vomiting and use of activated charcoal can reduce the absorption of drug. If

coma ensues, intensive care measures should be employed on an emergency basis.

The treatment of withdrawal syndrome is usually conservative. However, pentobarbital substitution therapy has been suggested for the treatment of withdrawal from short-acting barbiturates. After the detoxification phase is over, follow-up supportive treatment and treatment of associated psychiatric disorder(s), usually depression, are important steps to prevent relapses.

BENZODIAZEPINES AND OTHER SEDATIVE-HYPNOTIC USE DISORDER

Since the discovery of chlordiazepoxide in 1957 by Sternbach, benzodiazepines have replaced other sedative-hypnotics in the treatment of insomnia and anxiety. These are currently one of the most often prescribed drugs.

Benzodiazepines produce their effects by acting on the benzodiazepine receptors (GABA-benzodiazepine receptor complex), thereby indirectly increasing the action of GABA, the chief inhibitory neurotransmitter in the human brain.

Benzodiazepine (or sedative-hypnotic) use disorder can be either iatrogenic or originating with illicit drug use. Dependence, both physical and psychological, can occur and tolerance is usually moderate.

Intoxication and Complications

Acute intoxication resembles alcohol intoxication, while chronic intoxication causes tolerance, especially to the sedative and anticonvulsant actions. Excessive doses can lead to respiratory depression, coma and death while chronic use has been reported to cause amnesic syndrome.

Withdrawal Syndrome

A typical withdrawal syndrome, after cessation of prolonged use (more than 4-6 weeks) of

TABLE 5.2. FIRST RANK SYMPTOMS (SFRS) OF SCHIZOPHRENIA**A. Hallucinations**

1. Audible thoughts: Voices speaking out thoughts aloud or 'thought echo'.
2. Voices heard arguing: Two or more hallucinatory voices discussing the subject in third person.
3. Voices commenting on one's action.

B. Thought Alienation Phenomena

4. Thought withdrawal: Thoughts cease and subject experiences them as removed by an external force.
5. Thought insertion: Subject experiences thoughts imposed by some external force on his passive mind.
6. Thought diffusion or broadcasting: Subject experiences that his thoughts are escaping the confines of his self and are being experienced by others around.

C. Passivity Phenomena

7. 'Made' feelings or affect.
8. 'Made' impulses.
9. 'Made' volition or acts.

In 'made' affect, impulses and volitions, the subject experiences feelings, impulses or acts which are imposed by some external force. In 'made' volition, for example, one's own acts are experienced as being under the control of some external force, the subject being like a robot.

10. Somatic passivity: Bodily sensations, especially sensory symptoms, are experienced as imposed on body by some external force.

D. Delusional Perception

11. Delusional perception: Normal perception has a private and illogical meaning.

included delusions, hallucinations and negativism.

Kurt Schneider

Kurt Schneider (1959) described symptoms which, though not specific of schizophrenia, were of great help in making a clinical diagnosis of schizophrenia.

These are popularly called as *Schneider's first rank symptoms* of schizophrenia (FRS or SFRS) (Table 5.2). He also described the *second rank symptoms* of schizophrenia (which are considered less important for diagnosis of schizophrenia), like other forms of hallucinations, perplexity, and affect disturbances.

These symptoms (SFRS) have been described in some detail here as they have very often been used for the diagnosis of schizophrenia and have significantly influenced the diagnostic criteria and classification of schizophrenia and other related psychotic disorders. As mentioned earlier, SFRS are not specific for schizophrenia and may be seen in other psychiatric disorders like mood disorders and organic psychiatric disorders.

EPIDEMIOLOGY

According to the World (Mental) Health Report 2001, about 24 million people worldwide suffer from schizophrenia.

The point prevalence of schizophrenia is about 0.5-1%. Schizophrenia is prevalent across racial, socio-cultural and national boundaries, with a few exceptions in the prevalence rates in some isolated communities.

The incidence of schizophrenia is believed to be about 0.5 per 1000. The onset of schizophrenia occurs usually later in women and often runs a more benign course, as compared to men.

CLINICAL FEATURES

Schizophrenia is characterized by disturbances in thought and verbal behavior, perception, affect, motor behavior and relationship to the external world. The diagnosis is entirely clinical and is based on the following clinical features, none of which are pathognomonic alone.

I. Thought and Speech Disorders

Autistic thinking is one of the most classical features of schizophrenia. Here the thinking is

1. Paranoid Schizophrenia

Paranoid schizophrenia is characterized by the following clinical features, in addition to the general guidelines of schizophrenia described earlier:

1. Delusions of persecution, reference, grandeur (or 'grandiosity'), control, or infidelity (or 'jealousy'). The delusions are usually well-systematized (i.e. thematically well connected with each other).
2. The hallucinations usually have a persecutory or grandiose content.

TABLE 5.4. SOME IMPORTANT CLINICAL FEATURES OF RETARDED CATATONIA

- Mutism:** Complete absence of speech
- Rigidity:** Maintenance of a rigid posture against efforts to be moved
- Negativism:** An apparently motiveless resistance to all commands and attempts to be moved, or doing just the opposite
- Posturing:** Voluntary assumption of an inappropriate and often bizarre posture for long periods of time
- Stupor:** Akinesia (no movement) with mutism but with evidence of relative preservation of conscious awareness
- Echolalia:** Repetition, echo or mimicking of phrases or words heard
- Echopraxia:** Repetition, echo or mimicking of actions observed
- Waxy flexibility:** Parts of body can be placed in positions that will be maintained for long periods of time, even if very uncomfortable; flexible like wax
- Ambitendency:** Due to ambivalence, conflicting impulses and tentative actions are made, but no goal directed action occurs, e.g. on asking to take out tongue, tongue is slightly protruded but taken back again
- Other Signs:** Like *mannerisms*, *stereotypies* (verbal and behavioral), *automatic obedience* (commands are followed automatically, irrespective of their nature) and *verbigeration* (incomprehensible speech).

3. No prominent disturbances of affect, volition, speech, and/or motor behavior.

The personality deterioration in the paranoid subtype is much less than that seen in the other types of schizophrenias. The patients may be quite apprehensive (due to delusions and hallucinations), and appear evasive and guarded on the mental status examination.

The onset of paranoid schizophrenia is usually insidious, occurs later in life (i.e., late 3rd and early 4th decade) as compared to the other subtypes of schizophrenia. The course is usually progressive and complete recovery usually does not occur. There may be frequent remissions and relapses. At other times, the functional capability may be only slightly impaired. The differential diagnosis is from delusional (or paranoid) disorders and paranoid personality disorders.

II. Disorganized (or Hebephrenic) Schizophrenia

Disorganized schizophrenia is characterized by the following features, in addition to the general guidelines of schizophrenia described earlier:

1. Marked thought disorder, incoherence and severe loosening of associations. Delusions and hallucinations are fragmentary and changeable.
2. Emotional disturbances (inappropriate affect, blunted affect, or senseless giggling), mannerisms, 'mirror-gazing' (for long periods of time), disinhibited behavior, poor self care and hygiene, markedly impaired social and occupational functioning, extreme social withdrawal and other oddities of behavior.

ICD-10 (CDDG) recommends a period of 2 or 3 months of continuous observation for a confident diagnosis of disorganized (or hebephrenic) schizophrenia to be made.

The onset is insidious, usually in the early 2nd decade. The course is progressive and downhill. The recovery from the episode is

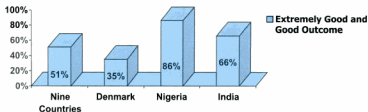


FIG. 5.1. IPSS OUTCOME

unmarried or widowed women. Delusions of persecution are seen, with bizarre and fantastic content (like being raped or gassed or strangers entering their rooms and interfering with them).

Hallucinations of all kinds (visual, auditory, tactile, gustatory and olfactory) can be present. Intelligence and social judgement outside the arena of persecutory delusions are usually normal. Approximately 25-40% of the patients have some defect of sight or hearing.

At present, this syndrome is placed under paranoid schizophrenia, late onset type.

F. Prolonged Schizophrenia

This is a syndrome of schizophrenia occurring in the presence of mental retardation. It differs from schizophrenia in only that there is often a poverty of ideation and delusions are not usually very well-systematized. Therefore, behavioral disturbances are much more prominent than delusions and hallucinations. However, both ICD-10 and DSM-IV-TR do not suggest a separate category and it is best to diagnose both schizophrenia and mental retardation, when present together.

G. Type I and Type II Schizophrenia

TJ Crow had divided schizophrenia into two sub-types, namely Type I and II schizophrenias. The Type I syndrome is characterized by positive

symptoms while the Type II syndrome is predominantly characterized by presence of negative symptoms.

Type I syndrome is supposed to have an acute presentation, good response to medication and a good outcome, while Type II is theorized to be chronic in course, have a poor response to medication and a poor outcome. Crow also described dilated ventricles on CT Scan of Brain in Type II syndrome.

Although a useful concept theoretically, it has not been found to be valid in the recent research studies. Very few patients have a pure Type I or Type II syndrome, and admixtures are far more common.

DIFFERENTIAL DIAGNOSIS

The first step in the differential diagnosis is to exclude psychoses with known organic causes, e.g. complex partial seizures, drug-induced psychoses (like amphetamine-induced psychoses), metabolic disturbances, or cerebral neoplasm.

The second step is to rule out a possibility of mood disorder (e.g. mania, depression, or mixed affective disorder) or schizo-affective disorder.

The third step is to exclude the possibility of other non-organic psychoses like delusional disorders, or acute and transient psychotic disorders (ATPD).

more than 35 hours contact per week with a 'significant other' with expressed emotions, were particularly likely to relapse (92%). The comparable rate in patients who received antipsychotic medication and faced less expressed emotions was 12%. In the research conducted in the last 30 years, these findings have been replicated all over the globe, though the period of 35 hours/week has not been found to be significant.

2. Family Theories

Several theories have been propounded in the past but are currently of doubtful value. These include 'schizophrenogenic mothers', lack of 'real' parents, dependency on mother, anxious mother, parental marital schism or skew, double-bind theory, communication deviance, and pseudo-mutuality.

3. Information Processing Hypothesis

Disturbances in attention, inability to maintain a set, and inability to assimilate and integrate percepts are common findings in schizophrenia. Patients with schizophrenia may at first be overly attentive to stimuli but later may reduce or exclude attention to stimuli. There is possibly a breakdown in the *internal representation* of mental events.

4. Psychoanalytical Theories

According to Freud, there is regression to the pre-oral (and oral) stage of psychosexual development, with the use of defense mechanisms of denial, projection, and reaction formation (Table 17.1). There is a loss of ego-boundaries (described by Federn), with a loss of touch with reality.

III. Socio-Cultural Theories

Although the prevalence of schizophrenia is quite uniform across cultures, it was found to be more common in lower socio-economic status in some studies. This has now been explained due

to a '*downward social drift*', which is a result of having developed schizophrenia rather than causing it.

Higher rates of schizophrenia have been found among some migrants, not only among the first generation migrants but also among the second generation.

MANAGEMENT

The treatment of schizophrenia can be discussed under the following headings:

1. Somatic treatment
 - a) Pharmacological treatment
 - b) Electro-convulsive therapy (ECT)
 - c) Others.
2. Psychosocial treatment and rehabilitation.

1. Pharmacological Treatment

The first drug to be used with beneficial effect in schizophrenia was reserpine (*Rauwolfia serpentina* extract), in India by Sen and Bose (1931). Reserpine is no longer used for the treatment of schizophrenia.

Antipsychotics were formally discovered by Delay and Deniker in 1952. Since their introduction, they have changed the outcome of schizophrenia significantly. These are discussed in some detail in Chapter 15. Only some relevant points are briefly mentioned here.

Some commonly used antipsychotics are listed in Table 5.7 with their routinely used dosage.

Atypical antipsychotic drugs, like risperidone, olanzapine, quetiapine, aripiprazole, and ziprasidone, are more commonly used than the typical antipsychotics like trifluoperazine and haloperidol, in acute stages. Atypical antipsychotics are also more useful when negative symptoms are prominent, e.g. in chronic schizophrenia.

Clozapine, another atypical antipsychotic, is available in the Indian market. The clinical trials have shown that clozapine is effective in about

6

Mood Disorders

Broadly speaking, the *emotions* can be described as two main types:

1. *Affect*, which is a *short-lived* emotional response to an idea or an event, and
2. *Mood*, which is a *sustained and pervasive* emotional response which colors the whole psychic life.

So according to these definitions, depression and mania are '*mood disorders*' and not '*affective disorders*' as they have been called so frequently in the past. Throughout this chapter (and this book), the more correct word '*mood disorder*' will be used (as indeed in DSM-IV-TR and ICD-10).

Mood disorders have been known to man since antiquity. The Old Testament describes King Saul as suffering from severe depressive episodes and responding slightly to David's soothing music. While Hippocrates coined the words mania and melancholia, it was Aretaeus who first described mania and depression occurring in the same individual.

Emil Kraepelin, borrowing from the work of Kahlbaum, Falret and Baillarger, described the manic-depressive illness as separate from dementia precox on the basis of course, clinical symptoms and outcome.

Recently, the World Health Report 2001 (WHO) has identified *unipolar depression* as the 4th cause of DALY (Disability-Adjusted Life Years) in all ages, and the 2nd cause in the age group of 15-44 years. Unipolar depression is also the 1st cause of YLD (Years of Life Lived with Disability) in all ages. The comparison was with

all medical disorders, and not only psychiatric disorders.

The World Health Report 2001 estimates that there are 121 million people worldwide suffering from depression.

CLASSIFICATION

The classification of mood disorders is an area which is fraught with multiple controversies. According to the ICD-10 (CDDG), the mood disorders are classified as follows:

1. Manic episode
2. Depressive episode
3. Bipolar mood (affective) disorder
4. Recurrent depressive disorder
5. Persistent mood disorder (including cyclothymia and dysthymia)
6. Other mood disorders (including mixed affective episode and recurrent brief depressive disorder).

CLINICAL FEATURES AND DIAGNOSIS

I. Manic Episode

The life-time risk of manic episode is about 0.8-1%. This disorder tends to occur in episodes lasting usually 3-4 months, followed by complete clinical recovery. The future episodes can be either manic, depressive or mixed.

A manic episode is typically characterized by the following features (which should last for at least one week and cause disruption in occupational and social activities):

of 6-12 months, after achieving full remission. It is wise to taper the antidepressant medication, when the treatment is to be stopped after the continuation phase.

There are three main phases of treatment:

- i. *Acute treatment* (till remission occurs),
- ii. *Continuation treatment* (from remission till end of treatment), and
- iii. *Maintenance treatment* (to prevent further recurrences).

Maintenance treatment may be indicated in the following patients:

- i. Partial response to acute treatment.
- ii. Poor symptom control during the continuation treatment.
- iii. More than 3 episodes (90% chances of recurrence).
- iv. More than 2 episodes with early age of onset, or recurrence within 2 years of stopping anti-depressants, or severe and / or life-threatening depression, or family history of mood disorder.
- v. Chronic depression (> 2 years) or double depression.

About 20-35% of depressed patients are *refractory* to antidepressant medication. The management of a treatment refractory depressed patient is best done by a psychiatrist, often at a tertiary care centre. These patients may require one of the following alternatives:

- i. A change of antidepressant (*Switch*),
- ii. Combination of two types of antidepressants,
- iii. Augmentation with lithium,
- iv. Augmentation with T_3 or T_4 ,
- v. Augmentation with antipsychotics,
- vi. Electroconvulsive therapy, or
- vii. Use of newer and experimental techniques.

One type of depression, namely delusional depression (depression with *psychotic* features),

is usually refractory to antidepressants alone. The treatments of choice in this condition include:

- i. An antidepressant with ECT, or
- ii. An antidepressant with antipsychotics, or
- iii. An antidepressant with lithium.

2. ECT (Electroconvulsive Therapy)

The indications for ECT in depression include:

- i. Severe depression with suicidal risk.
- ii. Severe depression with stupor, severe psychomotor retardation, or somatic syndrome.
- iii. Severe treatment refractory depression.
- iv. Delusional depression (psychotic features).
- v. Presence of significant antidepressant side-effects or intolerance to drugs.

Severe depression with suicidal risk is the first and foremost indication for use of ECT. The prompt use of ECT can be life-saving in such a situation.

The response is usually rapid, resulting in a marked improvement. In most clinical situations, usually 6-8 ECTs are needed, given three times a week. When six ECTs are administered, the usual pattern is three ECTs in the first week, two in the second week and one in the third week.

However, improvement is not sustained after stopping the ECTs. Therefore, antidepressants are often needed along with ECTs, in order to maintain the improvement achieved. The safety of the ECT procedure has now been well established.

ECT can also be used for acute manic excitement, if it is not adequately responding to antipsychotics and mood stabilizers.

3. Lithium (Li)

Lithium has traditionally been the drug of choice for the treatment of manic episode (acute phase) as well as for prevention of further episodes in bipolar mood disorder. It has also been used in

1. *Depression with predominant anxiety*: The anxiety here is far more subjective, in contrast to the objective restlessness seen in agitated depression. The treatment is usually by antidepressants.
2. *Phobic-anxiety-depersonalization syndrome (PAD syndrome)*: Described by Roth, this syndrome is commoner in women aged 20-40 years. It is characterized by diffuse anxiety and panic attacks, multiple phobias (like agoraphobia and claustrophobia), depersonalization, derealization and depressive features. The treatment is usually by antidepressants. It is recognized that ECTs may make the condition worse.
3. *Non-endogenous depression*: Atypical depression of the non-endogenous type is characterized by absence of neurotic traits and significant stressful life events. This differentiates it from neurotic or reactive depression.
4. *Hysteroid-dysphoric depression*: In this type, there are marked fluctuations of mood, ranging from near normal to severe depression. Any stress leads to an abrupt onset of depression, with marked anxiety, pallor and changes in physical appearance. The person usually has histrionic personality traits or disorder.

Atypical depression usually has an onset in the teens and runs a chronic course. Hypersomnia, hyperphagia, reactive nature of symptoms, 'rejection sensitivity' and lethargy ('leaden paralysis') are some of the characteristic features.

The importance of diagnosing atypical depression lies in the fact that many of these patients respond better to MAOIs (monoamine oxidase inhibitors), although SSRIs (Selective Serotonin Reuptake Inhibitors) and TCAs (Tricyclic Antidepressants) should be tried first.

Double Depression

This is a major depressive episode (usually acute), superimposed on an underlying dysthymia or

neurotic depression (usually chronic). The response to treatment is usually poor.

Agitated Depression

This is a type of severe depression with marked motor restlessness or agitation. It is either seen alone or along with involuntional melancholia. It is more common after the age of 40 years.

The treatment of agitated depression usually requires addition of antipsychotics or benzodiazepines to the antidepressant therapy.

Seasonal Mood Disorder

This is either a bipolar mood disorder or recurrent depressive episode which tends to occur in the same season on each occasion. It is usually more commonly seen in women.

For example, in a bipolar seasonal mood disorder, depression would tend to occur in the same months every time (usually winter months), while mania would occur in the months of some other season every episode (usually summer months).

This is thought to be due to changes in the length of the day (and light) and its effect on hypothalamus. The characteristic symptoms of winter depression are dysphoria, decreased activity and atypical depressive symptoms (increased fatigue, increased sleep, increased appetite and weight, and carbohydrate craving). The prevalence of winter seasonal mood disorder increases with increasing latitude.

The treatment, in addition to the usual modes of management, also includes *phototherapy*, which consists of increasing the number of hours of day-light (using artificial, full-spectrum, white light of 2,500-10,000 lux intensity, in the early morning, with eyes open) for the treatment of depression.

Secondary Depression and Secondary Mania

Depressive episode and manic episode can occur secondary to certain physical diseases and

This signal anxiety arouses the ego to take defensive action which is usually in the form of *repression*, a primary defense mechanism. Ordinarily when repression fails, other secondary defense mechanisms (like conversion, isolation) are called into action.

In anxiety, repression fails to function adequately but the secondary defense mechanisms are *not* activated. Hence, anxiety comes to the fore-front unopposed.

Developmentally, primitive anxiety is manifested as somatic symptomatology while developmentally advanced anxiety is signal anxiety.

Panic anxiety, according to this theory, is closely related to the separation anxiety of childhood.

2. Behavioral Theory

According to this theory, anxiety is viewed as an unconditioned inherent response of the organism to the painful or dangerous stimuli. In anxiety and phobias, this becomes attached to relatively neutral stimuli by conditioning.

The behavioral approach is generally more helpful in treatment rather than in explaining the cause of anxiety.

3. Cognitive Behavioral Theory (CBT)

According to cognitive behavior theory, in anxiety disorders, there is evidence of selective information processing (with more attention paid to threat-related information), cognitive distortions, negative automatic thoughts (NATs) and perception of decreased control over both internal and external stimuli.

4. Biological Theory

i. Genetic evidence

About 15-20% of first degree relatives of the patients with anxiety disorder exhibit anxiety disorders themselves. The concordance rate in the monozygotic twins of patients with panic disorders is as high as 80% (4 times more than in dizygotic twins).

ii. Chemically induced anxiety states

Infusion of chemicals (like sodium lactate, isoproterenol and caffeine), ingestion of yohimbine and inhalation of 5% CO₂ can produce panic episodes in predisposed individuals. Administration (oral) of MAOIs before the lactate infusion protects the individual(s) from panic attack, thus providing a probable clue to the biological model of anxiety.

iii. GABA-benzodiazepine receptors

This is one of the most recent advances in the search for the etiology of anxiety disorders. The benzodiazepine receptors are distributed widely in the central nervous system. Presently, two types of benzodiazepine receptors have been identified. The type I (ω_1) is GABA and chloride independent, while type II (ω_2) is GABA and chloride dependent.

GABA (Gamma amino butyric acid) is the most prevalent inhibitory neurotransmitter in the central nervous system. It has been suggested that an alteration in GABA levels may lead to production of clinical anxiety. The fact that the benzodiazepines (which facilitate GABA transmission, thereby causing a generalized inhibitory effect on the CNS) relieve anxiety and that benzodiazepine-antagonists (e.g. flumazenil) and inverse agonists (e.g. β -carbolines) cause anxiety, lends heavy support to this hypothesis.

iv. Other neurotransmitters

Norepinephrine, 5-HT, dopamine, opioid receptors and neuroendocrine dysfunction have also been implicated in the causation of anxiety disorders.

v. Neuroanatomical basis

Locus ceruleus, limbic system, and prefrontal cortex are some of the areas implicated in the etiology of anxiety disorders. Regional cerebral blood flow (rCBF) is increased in anxiety, though vasoconstriction occurs in severe anxiety.

Differential Diagnosis

The differential diagnoses include anxiety disorder, panic disorder, major depression, avoidant personality disorder, obsessive compulsive disorder, delusional disorder, hypochondriasis, and schizophrenia.

Treatment

Most patients with phobic disorder rely on avoidance to manage their fears and anxieties. As long as they find ways to limit their lives within the limitations imposed by phobias, they experience little, if any, anxiety. When they are forced to face the phobic situation, anxiety mounts and they then seek treatment. The patients with more than one phobia and presence of panic symptoms often seek treatment earlier.

The treatment approach is usually multimodal.

1. Psychotherapy

Psychodynamically oriented psychotherapy is not usually helpful in treatment of phobias. This approach is however indicated when there are characterological or personality difficulties as well.

Supportive psychotherapy is a helpful adjunct to behavior therapy and drug treatment.

As stated earlier, *Cognitive behavior therapy* (CBT) can be used to break the anxiety patterns in phobic disorder. It is usual to combine CBT with behavioral techniques.

2. Behavior Therapy

If properly planned, this mode of treatment is usually successful. The behavioral therapies are discussed in Chapter 18 and only the names of important techniques are mentioned here.

- i. Flooding.
- ii. Systematic desensitization.
- iii. Exposure and response prevention.
- iv. Relaxation techniques.

3. Drug Treatment

The drugs used in the treatment of phobia are:

- i. Benzodiazepines (Chapter 15) are useful in reducing the anticipatory anxiety.

Alprazolam is stated to have anti-phobic, anti-panic and anti-anxiety properties. So, it is the drug of choice, when benzodiazepines are used. However, long-term, double-blind randomized controlled trials are needed. The other drugs used include clonazepam and diazepam.

However, long-term use of benzodiazepines is fraught with the dangers of tolerance and dependence.

- ii. Among the antidepressants (discussed in Chapter 15), SSRIs are currently drugs of choice, with paroxetine being the most widely used drug. Other SSRIs, like fluoxetine and sertraline are also equally effective. Fluoxetine has the advantage of a longer half-life.

Other antidepressants, e.g. imipramine (TCA) and phenelzine (MAOI), are also helpful in treating the panic attacks associated with phobias, thereby decreasing the distress.

As mentioned earlier, multiple approaches are usually combined together in the treatment of a particular patient.

OBSESSIVE-COMPULSIVE DISORDER

An *obsession* is defined as:

1. An idea, impulse or image which intrudes into the conscious awareness repeatedly.
2. It is recognized as one's own idea, impulse or image but is perceived as *egoalien* (foreign to one's personality).
3. It is recognized as irrational and absurd (insight is present).
4. Patient tries to resist against it but is unable to.
5. Failure to resist, leads to marked distress.

- ii. Supportive psychotherapy is an important adjunct to other modes of treatment. Supportive psychotherapy is also needed by the family members.

2. Behavior Therapy and Cognitive Behavior Therapy

Behavior modification is an effective mode of therapy, with a success rate as high as 80%, especially for the compulsive acts. It is customary these days to combine CBT with BT at most centres.

The techniques used are listed below (They are described in Chapter 18):

- i. Thought-stopping (and its modifications).
- ii. Response prevention.
- iii. Systematic desensitization.
- iv. Modeling.

3. Drug Treatment

- i. Benzodiazepines (e.g. alprazolam, clonazepam) have a limited role in controlling anxiety as adjuncts and should be used very sparingly.
- ii. Antidepressants: Some patients may improve dramatically with specific serotonin reuptake inhibitors (SSRIs).

Clomipramine (75-300 mg/day), a non-specific serotonin reuptake inhibitor (SRI), was the first drug used effectively in the treatment of OCD. The response is better in the presence of depressive symptoms, but many patients with pure OCD also improve substantially.

Fluoxetine (20-80 mg/day) is a good alternative to clomipramine and often preferred these days for its better side-effect profile. *Fluvoxamine* (50-200 mg/day) is marketed as a specific anti-obsessional SSRI drug, while *paroxetine* (20-40 mg/day), and *sertraline* (50-200 mg/day) are also effective in some patients.

- iii. *Antipsychotics*: These are occasionally used in low doses (e.g. haloperidol, risperidone, olanzapine, aripiprazole pimozide) in the treatment of severe, disabling anxiety.
- iv. *Bupirone* has also been used beneficially as an adjunct for augmentation of SSRIs, in some patients.

4. Electroconvulsive Therapy

In the presence of severe depression with OCD, ECT may be needed. ECT is particularly indicated when there is a risk of suicide and/or when there is a poor response to the other modes of treatment. However, ECT is not the treatment of first choice in OCD.

5. Psychosurgery

Psychosurgery can be used in treatment of OCD that has become intractable, and is not responding to other methods of treatment. It is worth mentioning that psychosurgery is only available as a treatment choice at a very few centres throughout the world.

The best responders are usually those who have significant associated depression, although pure obsessives also do respond. The main benefit is the marked reduction in associated distress and severe anxiety.

The procedures which can be employed are:

- i. Stereotactic limbic leucotomy.
- ii. Stereotactic subcaudate tractotomy.

Psychosurgery is usually followed by intensive behavior therapy aimed at rehabilitation. However, with the easy availability of SSRIs, and a good response of OCD symptoms to SSRIs and other pharmacological measures, psychosurgery is very rarely used in the treatment of OCD.

Very often, a comprehensive treatment of obsessive-compulsive disorder requires that multiple treatment modalities (e.g. drug treatment and BT/CBT) be combined in a specific manner, suitable to the particular patient being treated at the time.

TABLE 8.5. TYPES OF DISSOCIATIVE AMNESIA

- 1. Circumscribed Amnesia** (commonest type):
There is an inability to recall all the personal events during a circumscribed period of time, usually corresponding with the presence of the stressor.
- 2. Selective Amnesia** (less common):
This is similar to circumscribed amnesia but there is an inability to recall only some selective personal events during that period while some other events during the same period may be recalled.
- 3. Continuous Amnesia** (rare):
In this type, there is an inability to recall all personal events following the stressful event, till the present time.
- 4. Generalized Amnesia** (very rare):
In this type, there is an inability to recall the personal events of the whole life, in the face of a stressful life event.

clouding of consciousness. In the post-amnesic period, the awareness of disturbance of memory is present.

2. Dissociative Fugue

Dissociative fugue is characterized by episodes of wandering away (usually away from home). During the episode, the person usually adopts a new identity with complete amnesia for the earlier life. The onset is usually sudden, often in the presence of severe stress. The termination too is abrupt and is followed by amnesia for the episode, but with recovery of memories of earlier life. The characteristic feature is the assumption of a purposeful new identity, with absence of awareness of amnesia.

An important differential diagnosis is from fugue states seen in complex partial seizures or temporal lobe epilepsy. In complex partial seizures, there is no assumption of a new identity, confusion or disorientation is present during the episode and the episodes are not only linked to any precipitating stress.

3. Multiple Personality (Dissociative Identity) Disorder

In this dissociative disorder, the person is dominated by two or more personalities, of which only one is being manifest at a time. These personalities are usually different, at times even opposing. Each personality has a full range of higher mental functions, and performs complex behavior patterns.

Usually one personality is not aware of the existence of the other(s), i.e. there are amnesic barriers between the personalities. Both the onset and termination of control of the each personality is sudden.

Classical examples in the published literature include 'Three faces of Eve' and 'Sybil'.

4. Trance and Possession Disorders

Trance and possession disorders (possession hysteria) are characterized by the control of person's personality by a 'spirit', during the episodes. Usually the person is aware of the existence of the other (i.e. 'possessor'), unlike in multiple personality. This disorder is very commonly seen in India and certain African countries.

5. Other Dissociative Disorders

Ganser's syndrome (hysterical pseudodementia) is commonly found in prison inmates. The characteristic feature is *vorbeireden*, which is also called as 'approximate answers'. The answers are wrong but show that the person understands the nature of question asked. For example; when asked the color of a red pen, the patient calls it blue.

Etiology

The etiological theories of dissociative (and conversion) disorders are predominantly of three types:

1. Psychodynamic Theory

The explanation given by this theory can be summarized in a flow diagram (Fig 8.2).

term depersonalization disorder should be used only when other disorders have been ruled out.

The other associated clinical features include:

1. The episodes of depersonalization and/or derealization causing significant social, interpersonal or occupational impairment. The episodes recur frequently.
2. The onset and termination of episodes is usually sudden.
3. Marked distress and anxiety results, as the experience is highly unpleasant. Insight into the illness is usually present.
4. Feeling of loss of control on one's action and speech may occur.
5. The episodes occur in the presence of a clear sensorium.

The age of onset is usually late second or early third decade. The course is usually chronic.

Treatment

The treatment is usually not very successful though co-morbid symptoms of anxiety and depression can often be treated. The various methods which can be tried include:

1. Supportive psychotherapy.
2. Drug therapy with antidepressants; rarely amphetamines or antipsychotics are also tried.

3. Other Specified Neurotic Disorders (Culture Bound Syndromes)

In ICD-10, this category includes miscellaneous disorders which are of uncertain etiology and nosological status, and which occur in certain cultures, e.g. *dhat syndrome*, *koro*, *latah*, *wihtigo*, *piblokto* and *amok*. These are called as *culture-bound syndromes*, as they are localized to certain geographical areas only, and are not seen in the Western, developed countries.

A. Dhat Syndrome

Dhat syndrome is a culture-bound syndrome, which is prevalent in the Indian subcontinent. This is characterized by:

1. Complaint of passage of '*dhat*' in urine.
2. Multiple somatic symptoms.
3. Asthenia (physical or mental exhaustion).
4. Anxiety or depression may be present.
5. Sexual dysfunction may occur.

'Dhat' is a whitish discharge passed in urine and believed to be semen by the patient. According to a socio-cultural belief prevalent in the Indian society, semen is an 'extremely precious' body element which is produced from many drops of blood. Hence, it follows from this view that loss of semen will be perceived to lead to weakness and sexual dysfunction. Often, masturbatory anxiety and overconcern with nocturnal emissions are also associated with other clinical symptoms.

Treatment

1. *Counseling*: This is the most important method of treatment directed towards removing the misconceptions regarding apprehension of semen loss. This counseling is combined with general sex education. Specifically, fears regarding masturbation and nocturnal emissions are allayed.
2. *Symptomatic treatment*: The treatment of underlying anxiety, depression, hypochondriasis and/or sexual dysfunction by the usual means may also be required.

B. Amok

Amok is characterized by a sudden, unprovoked episode of rage, in which the affected person runs about (runs '*amok*') and indiscriminately injures or kills any person who is encountered on the way. This condition is usually seen in south-east Asia (e.g. Malaysia).

C. Koro

Koro is a culture-bound syndrome seen in Asia (including India). The affected male person has the belief that his penis is shrinking and may disappear in to his abdominal wall and he may then die.

The life-time prevalence of personality disorders in the general population is about 5-10%. Often symptoms of more than one personality disorder are present in one person. In fact, it is now believed that the occurrence of mixed personality disorders (i.e. comorbidity) is commoner than single (pure) personality disorders.

In DSM-IV-TR, the personality disorders (and traits) are coded on Axis II (on the multi-axial system) (see Chapter 1) and have been divided into three clusters (Table 9.1).

In addition to these three clusters, some other personality disorders like passive-aggressive personality disorder and depressive personality disorder have been included in the section on 'Criteria provided for further study' in DSM-IV-TR.

In ICD-10, the personality disorders are listed under the section on 'Disorders of adult personality and behavior'.

Diagnosis

According to the ICD-10 (CDDG), the diagnostic guidelines for *specific personality disorder* include conditions not directly attributable to gross brain damage or disease, or to another psychiatric disorder, meeting the following criteria:

1. Markedly disharmonious attitudes and behavior, involving usually several areas of functioning, e.g. affectivity, arousal, impulse control, ways of perceiving and thinking, and style of relating to others;
2. The abnormal behavior pattern is enduring, of long standing, and not limited to episodes of mental illness;
3. The abnormal behavior pattern is pervasive and clearly maladaptive to a broad range of personal and social situations;
4. The above manifestations always appear during childhood or adolescence and continue into adulthood;

5. The disorder leads to considerable personal distress but this may only become apparent late in its course;
6. The disorder is usually, but not invariably, associated with significant problems in occupational and social performance.

Clinical Subtypes

1. Paranoid Personality Disorder

According to the ICD-10 (CDDG), the diagnostic guidelines for paranoid personality disorder include the following features (in addition to features of personality disorders in general, described above). A clear evidence is usually required of the presence of *at least three* of the traits or behaviors given in the clinical description.

1. Excessive sensitiveness to setbacks and rebuffs.
2. Tendency to bear grudges persistently, i.e. refusal to forgive insults and injuries or slights.
3. Suspiciousness and a pervasive tendency to distort experience by misconstruing the neutral or friendly actions of others as hostile or contemptuous.
4. A combative and tenacious sense of personal 'right', out of keeping with the actual situation.
5. Recurrent suspicions, without justification, regarding sexual fidelity of spouse or sexual partner.
6. Tendency to experience excessive self-importance, manifest in a persistent self-referential attitude.
7. Preoccupation with unsubstantiated 'conspiratorial' explanations of events, both immediate to the patient and in the world at large.

The patients may become involved in litigation on small issues. The disorder is commoner in men, and it is more common in minority groups and immigrants.

6. Narcissistic Personality Disorder

A relatively new concept in classification, this disorder is characterized by:

1. Ideas of grandiosity and inflated sense of self-importance.
2. Preoccupation with fantasies of unlimited success.
3. Attention seeking, dramatic behavior, needs constant praise, and unable to face criticism.
4. Lack of empathy with others, with exploitative behavior.
5. Shaky self-esteem, underlying sense of inferiority, easily depressed by minor events.

Treatment

Individual psychotherapy that is psychoanalytically oriented probably helps, but data available is not sufficient for further comment.

7. Emotionally Unstable Personality Disorder

According to the ICD-10 (CDDG), emotionally unstable personality disorder is described as a disorder in which there is a marked tendency to act impulsively without consideration of the consequences, together with affective instability. This disorder is further classified in to two types: impulsive type and borderline type.

The *impulsive type* is characterized by emotional instability and lack of impulse control. Outbursts of violence or threatening behavior are common, particularly in response to criticism by others.

The *borderline type* is characterized by emotional instability. In addition, patient's own self-image, aims, and internal preferences (including sexual) are often unclear or disturbed. There are usually chronic feelings of emptiness. A liability to become involved in intense and unstable relationships may cause repeated emotional crises and may be associated with excessive efforts to avoid abandonment and a series of suicidal threats or acts of self-harm, (although these may occur without obvious precipitants).

The borderline type is also known as *borderline personality disorder* (DSM-IV-TR), the characteristic features of which include the following:

1. Significant and persistent disturbance of identity of self, e.g. 'who am I'. There is marked uncertainty about major issues in life.
2. Unstable and intense interpersonal relationship patterns.
3. Impulsivity.
4. Unstable emotional responses, with rapid shifts. Anger outbursts may occur.
5. Chronic feelings of boredom or emptiness with inability to stay alone.
6. Deliberate self harm is common in the form of self-mutilation, suicidal gestures, or accident-proneness.

The term *borderline personality disorder* currently includes *ambulatory schizophrenia* and *pseudoneurotic schizophrenia*, which were earlier thought to be subtypes of schizophrenia.

Psychodynamically, *splitting* is the primary defense mechanism employed in *borderline personality disorder*.

There is a considerable overlap between *borderline*, *narcissistic* and *antisocial (dissocial)* personality disorders (they belong to the same Cluster of personality disorders, i.e., B). Major depressive episodes occur commonly in this disorder.

Treatment

1. Psychoanalysis or Psychoanalytical psychotherapy
2. Supportive psychotherapy.
3. Drug therapy: Antidepressants have been used with success in certain patients with depression. Major depressive episode, if occurs, necessitates antidepressant therapy. Occasionally antipsychotics, lithium, and carbamazepine have been used when aggression and impulsivity are prominent.

Drug therapy is not the treatment of first choice in *borderline personality disorder*.

10

Sexual Disorders

The sexual disorders can be classified into four main types:

1. Gender identity disorders.
2. Psychological and behavioral disorders associated with sexual development and maturation.
2. Paraphilias (disorders of sexual preference).
3. Sexual dysfunctions.

In ICD-10, gender identity disorders, disorders of sexual preference, and sexual development and orientation disorders are listed under the disorders of adult personality and behavior, while sexual dysfunctions (not caused by organic disorder or disease) are listed under the behavioral syndromes associated with physiological disturbances and physical factors.

GENDER IDENTITY DISORDERS

These disorders are characterized by disturbance in *gender identity*, i.e. the sense of one's masculinity or femininity is disturbed. This group includes:

- A. Transsexualism: male and female; primary and secondary.
- B. Gender identity disorder of childhood.
- C. Dual-role transvestism.
- D. Intersexuality.

A. Transsexualism

Transsexualism, the severest form of gender identity disorders, is characterized by the following clinical features:

1. Normal anatomic sex.
2. Persistent and significant sense of discomfort regarding one's anatomic sex and a feeling that it is inappropriate to one's perceived-gender.
3. Marked preoccupation with the wish to get rid of one's genitals and secondary sex characteristics, and to adopt the sex characteristics of the other sex (perceived-gender).
4. Diagnosis is made after puberty.

Transsexualism is of two main types: primary and secondary.

1. Primary Transsexualism

This condition has an onset in early childhood and has a stable course over time. This is a relatively homogenous category. Primary transsexuals, more than secondary transsexuals, are preoccupied with sex-change or sex-reassignment surgery. There are two main types:

- i. male (or, male-to-female) primary transsexualism.
- ii. female (or, female-to-male) primary transsexualism.

2. Secondary Transsexualism

In contrast to the primary type, secondary transsexualism usually has an onset later in life. This is a less severe and more heterogeneous category. The only common feature is a wish to change the anatomic sex.

This category includes effeminate homosexuals, transvestites who secondarily become

The causes can be biological (e.g. endocrinal disorders like hypothyroidism, drug-induced) or psychological (e.g. marital conflicts).

8. Premature ejaculation

This disorder is defined as ejaculation before the completion of *satisfactory* sexual activity for both partners. In severe cases, it is characterized by ejaculation either before penile entry into vagina or soon after penetration. It is a very common disorder in the clinical setting.

The causes can be biological (relatively uncommon) or psychological (e.g. performance anxiety).

IV. Sexual Pain Disorders

9. Non-organic vaginismus

This disorder is characterized by an involuntary spasm of the lower 1/3rd of vagina, interfering with coitus. Penile entry is either painful or impossible.

Before a presumption of non-organic vaginismus, it is important to rule out organic factors (e.g. local pathology causing pain).

10. Non-organic dyspareunia

This disorder is characterized by pain in the genital area of either male or female, during coitus.

Before a presumption of non-organic dyspareunia, it is particularly important to rule out organic factors (e.g. local pathology causing pain) in both males and females.

V. Sexual Disorders Due to A General Medical Condition

The disorders listed above may occur secondary to a general medical condition; they should then be coded here.

These dysfunctions are called disorders only when they occur recurrently and persistently. Also if the sexual dysfunction is due to sexual stimulation being inadequate either in focus,

intensity or duration, a diagnosis of excitement and orgasmic phase disorders is not made.

Diagnosis and Differential Diagnosis

Before making a diagnosis of sexual dysfunction, it is of paramount importance to rule out an underlying physical cause, which would need treatment.

Although the diagnosis is clinical, a detailed physical examination and laboratory investigations (e.g. blood counts, blood sugar, liver function tests, thyroid function tests, hormonal profile, and rarely, routine examination and culture of prostatic fluid) coupled with a good history is a must in every patient to rule out an underlying physical cause.

Certain laboratory techniques (e.g. penile plethysmograph, penile tumescence monitoring during sleep) may help in differentiating organic and non-organic sexual dysfunctions. If NPT (nocturnal penile tumescence) is abnormal, then ancillary investigations like penile vascular investigations (e.g. penile pulse pressure, penile Doppler, duplex ultrasonography, diagnostic intracavernosal vasoactive substance-papaverine injection test, arteriography, DICC-dynamic infusion cavernosomatogram and cavernosogram, and cavernosography) and penile neurological investigations (e.g. penile sensory threshold test or penile biothesiometry) may be employed.

Though searching for an organic factor responsible for the sexual dysfunction is very important, a large majority of dysfunctions are psychosexual in nature. A detailed sexual and personal history is important in finding out the underlying causes. The common psychological causes of sexual dysfunction have been discussed earlier.

It should be specified whether the sexual dysfunction is psychogenic alone or biogenic factors co-exist; whether the dysfunction is life-long or acquired; and whether the dysfunction is situational or generalized.

11

Sleep Disorders

SLEEP

Nearly one third of our life is spent in sleep. Every day we go to our beds and pass into an easily reversible state of relative unresponsiveness and serenity. This state occurs more or less regularly and repetitively each day.

The EEG recordings show typical features of sleep. It is known that sleep is basically divided into two broadly different phases:

1. *D-sleep* (desynchronized or dreaming sleep), also called as *REM-sleep* (rapid eye movement sleep), or *active sleep*, or *paradoxical sleep*.
2. *S-sleep* (synchronized sleep), also called as *NREM-sleep* (non-REM sleep), or *quiet sleep*, or *orthodox sleep*.

S-sleep or NREM-sleep is further divided into 4 stages, ranging from stage 1 to stage 4. As the person falls asleep, the person first passes through these stages of NREM-sleep.

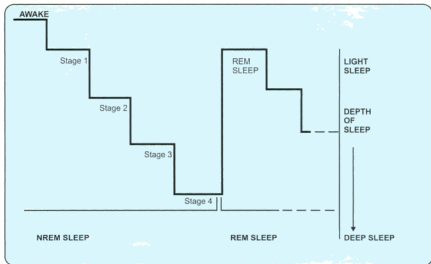


FIG. 11.1. STAGES OF SLEEP

II. Parasomnias

Parasomnias are dysfunctions or episodic nocturnal events occurring with sleep, sleep stages or partial arousals. Most parasomnias are common in the childhood though they may persist into adulthood.

A. Stage 4 Sleep Disorders

These are disorders occurring during deep sleep, i.e. Stages 3 and 4 of NREM sleep. The common Stage 4 parasomnias are:

1. *Sleep-walking (somnambulism)*: The patient carries out automatic motor activities that range from simple to complex. He may leave the bed, walk about or leave the house. Arousal is difficult and accidents may occur during sleep-walking.
2. *Sleep-terrors or night terrors (pavor nocturnus)*: The patient suddenly gets up screaming, with autonomic arousal (tachycardia, sweating, and hyperventilation). He may be difficult to arouse and rarely recalls the episode on awakening. In contrast, nightmares (which occur during REM sleep) are clearly remembered in the morning.
3. *Sleep-related enuresis (bedwetting)*: This is discussed in detail in Chapter 14.
4. *Bruxism (teeth-grinding)*: The patient has an involuntary and forceful grinding of teeth during sleep. Though the bed partner reports the loud sound produced by the grinding of teeth, and the destruction of the tooth enamel is obvious, the patient remains completely unaware of the episode.
5. *Sleep-talking (somniloquy)*: The patient talks during the Stages 3 and 4 of sleep but does not remember anything about it in the morning on awakening.

These disorders are often coexistent. As they occur during stage 4 (and 3) of NREM sleep, they are more common during the first one-third of the night (There is more NREM sleep in the first third of the night while the last third has more REM sleep). Arousal is difficult and on waking up, there is a complete amnesia for the event(s).

Treatment

Since benzodiazepines suppress Stage 4 of NREM sleep, a single dose at bedtime usually provides relief from Stage 4 parasomnias.

B. Other Sleep Disorders

Nightmares (dream anxiety disorder) occur during the REM sleep. They are characterized by fearful dreams occurring most commonly in the last one-third of night sleep. The person wakes up very frightened and remembers the dream vividly.

This is in contrast to *night terrors* which occur early in the night, are a Stage 4 NREM disorder, and are characterized by complete amnesia. In both the conditions, the observer finds the person frightened during the episode.

Other sleep disorders include nocturnal angina, nocturnal asthma, nocturnal seizures, paroxysmal nocturnal hemoglobinuria, nocturnal head banging, and familial sleep paralysis.

Treatment

There is no specific treatment. Treatment of the underlying condition is the most important step. The treatment of nightmares is by suppression of REM sleep e.g. by bedtime dose of a benzodiazepine. However on stopping the drug, a rebound increase in symptoms may occur.

2. Vomiting often occurs in complete absence of nausea or retching (Patients say that food just seems to come back up).
3. Vomiting is often self-induced and can be suppressed, if necessary.
4. Despite repeated vomiting, the weight loss is not significant.
5. The course of the illness is usually chronic with frequent remissions and relapses.

Treatment

1. Identification of psychosocial stressor.
2. Environmental manipulation to deal with stress.
3. Teaching coping skills.
4. Psychotherapy.

NON-ORGANIC SLEEP DISORDERS

The non-organic sleep disorders are discussed in Chapter 11.

SEXUAL DYSFUNCTIONS, NOT CAUSED BY ORGANIC DISORDERS OR DISEASE

The non-organic sexual dysfunctions are discussed in Chapter 10.

POST PARTUM PSYCHIATRIC DISORDERS (MENTAL AND BEHAVIORAL DISORDERS ASSOCIATED WITH PUERPERIUM, NOT CLASSIFIED ELSEWHERE)

Pregnancy and puerperium are highly stressful periods in a woman's life. The person is threatened by:

1. Physical changes, physiological changes and endocrinal changes occurring in one's body,
2. Reorganization of psyche in accordance with the new 'mother-role' (especially in the first pregnancy),
3. Body image changes, and
4. Unconscious intrapsychic conflicts relating to pregnancy, childbirth and motherhood become activated.

It is no wonder then that 25-50% of all women develop mild psychological symptoms in the puerperal period. The commonest type of picture is mild depression and irritability, known as *post-natal blues*. These 'blues' pass off within a few days and usually need no active management.

But, in 1.5-4.6/1,000 deliveries (average 2/1000 deliveries), the mother develops severe psychiatric symptoms or becomes psychotic. This is known as *post partum psychosis* (PPP).

The post partum psychosis can be:

1. Depressive episode with psychotic symptoms (most common), or
2. Schizophrenia, or
3. Manic episode, or
4. Delirium (least common).

Typically, the onset of symptoms occurs 3-7 days after delivery. An onset before the 3rd day postpartum is rare. The symptoms usually peak by the end of 4th week, and often necessitate admission in a hospital.

As the puerperium lasts for 6 weeks, the relevant postpartum period for psychiatric symptomatology to appear is 6 weeks (according to ICD-10). Hence, any psychiatric symptoms appearing within 6-week period after delivery are called as *post partum psychiatric disorder(s)*, if they do not fulfill the criteria of the major psychiatric disorders.

It was earlier believed that schizophreniform disorder is the commonest PPP in the developing countries (like India), in contrast to the developed countries where depression is more common. However, recent studies negate this hypothesis. Even in India, depressive episode is the most common type of PPP encountered in routine clinical practice though psychotic symptoms are frequently present.

The factors involved in causation are biological (changes in endocrine status especially steroid withdrawal, sleep disturbances, amine

13

Mental Retardation

NORMAL CHILD DEVELOPMENT

For an understanding of the behavioral and psychological problems of childhood, it is essential to know the normal patterns of child development. Although no two children are alike, there are general similarities in the mental and physical development of all normal children. A newborn human infant is probably the most helpless of all mammalian infants, and needs much more time to become self-dependent.

The normal development of a child can be divided into four major areas (these are modified after Denver Development Screening Test, DDST).

1. Motor behavior.
2. Adaptive behavior.
3. Language.
4. Personal and social behavior.

As normal children cross these milestones or developmental levels at nearly expected age limits (within a few months' range), it is best to describe these developmental changes as *milestones*.

In addition to these milestones, there are other developmental parameters like height, weight, activity level and general health which have an important bearing on the development of a child.

As mentioned earlier, slight delay from these milestone dates is not abnormal. Statistically only those values, which are beyond two standard deviation from the mean, are abnormal in a normal population. However, statistical abnormality is not always the same as the clinical abnormality.

INTELLECTUAL DEVELOPMENT

Intellectual development goes hand-in-hand with the development of physical and behavioral characteristics. It can be divided into the following stages, according to Jean Piaget:

1. Sensori-Motor Stage

This stage extends from birth to 2 years of age, and is characterized by:

- i. Actions related to sucking, orality and assimilation of objects.
- ii. Ability to think of only one thought at a time.
- iii. Inanimate objects are given human qualities.
- iv. 'Out of sight' means ceasing to exist.

2. Concrete Thinking Stage

This stage lasts from 2 years to 7 years of life, and is characterized by:

- i. Egocentric thought with a unique logic of its own, involving a limited point of view and lacking introspection.
- ii. Inability to generalize from specific events and to specify from general events.

3. Abstract or Conceptual Thinking Stage

This stage begins at 7 years of age and lasts till 11 years of age, although it may be said to continue throughout life. This is characterized by:

- i. Ability to focus on several dimensions of a problem at one time, mentally.

14

Child Psychiatry

CLASSIFICATION IN CHILD PSYCHIATRY

The various psychiatric disorders seen in childhood (*'disorders of psychological development'* and *'behavioral and emotional disorders with onset usually occurring in childhood and adolescence'*) are discussed under the following headings:

1. Mental retardation
2. Specific developmental disorders
3. Pervasive developmental disorders
4. Hyperkinetic disorders
5. Conduct disorders
6. Tic disorders
7. Enuresis and encopresis
8. Speech disorders
9. Habit disorders
10. Other disorders.

MENTAL RETARDATION

Mental retardation (or learning disability) has been discussed in Chapter 13.

SPECIFIC DEVELOPMENTAL DISORDERS

Mental retardation is a generalized impairment in nearly all areas of functioning. In contrast, the specific developmental disorders (SDD) are characterized by an inadequate development in usually one specific area of functioning.

The deficit in functioning may be in scholastic skills, speech and language, and motor skills. These may include reading (developmental reading disorder), language (developmental

language disorder), arithmetic or mathematics (developmental arithmetic or mathematics disorder), articulation (developmental articulation disorder or phonological disorder), or coordination (developmental co-ordination disorder). Sometimes, more than one developmental disorder is present.

All the developmental disorders either cause an impairment in academic functioning at school, especially when language is affected, or an impairment in the daily activities.

A large majority of these children have an underlying cerebral disorder. Boys are usually more affected than girls.

Before making a diagnosis of SDD, it is important to keep in mind the mental age, IQ, socio-cultural background, schooling, impairment(s) in vision and hearing, or any neurological deficit.

1. Specific Reading Disorder

It is also called as developmental reading disorder or *dyslexia*.

The child presents with a serious delay in learning to read, that is evident from the early years. The problems may include omissions, distortions, or substitutions of words, long hesitations, reversal of words, or simply slow reading.

Writing and spelling are also impaired. It is important to differentiate the disorder from scholastic backwardness; therefore a proper assessment is mandatory.

Treatment

The best treatment of encopresis is *preventive*. The toilet training period should be made as consistent and smooth as possible. The family environment should be warm and understanding. The emotional disturbances of the child should not be ignored and should be dealt with at the earliest. The communication between the family members should be direct.

After encopresis has developed, the treatment of choice is *behavior therapy*, using reinforcements (both positive and negative). The other treatments include psychotherapy, biofeedback and imipramine (in non-retentive encopresis).

SPEECH DISORDERS

Stuttering is a disorder of speech, characterized by the following features:

1. Disturbed fluency and rhythm of speech.
2. Intermittent blocking.
3. Repetition of words rapidly.
4. Prolongation of sounds.
5. Associated anxiety or distress.

Earlier *stammering* and *stuttering* were differentiated on the basis of minor differences. At present, they are thought to be synonymous.

The disorder is a fairly common one, affecting 2-5% of all children and 0.5-1% of all adults. Males are more commonly (about three times) affected.

Differential diagnosis is from *cluttering*, which is characterized by an erratic and dysarrhythmic pattern of speech, with jerky and rapid spurting of words. Unlike in stuttering where the person is aware of the difficulty in speech, the affected person in cluttering is usually unaware of the abnormal speech pattern.

Treatment

The treatment is by behavior modification techniques like desensitization, biofeedback and stammer suppresser; and by techniques to

diminish anxiety like relaxation therapy, drug therapy, individual or group psychotherapy.

HABIT DISORDERS

These are stereotyped disorders which are intentionally and repetitively produced but serve no constructive or socially acceptable function.

The common habit disorders include thumb sucking, nail biting, pulling out of hair (trichotillomania), head banging, masturbation, teeth grinding, picking of nose, biting parts of the body, skin-scratching, body rocking, breath-holding, and swallowing of air (aerophagia).

These habits range from normal to abnormal, depending on the severity of occurrence and the time of presentation during the developmental period (what is normal in infancy, may be abnormal in later childhood).

Many of the habit disorders, particularly those which are self-stimulating in nature, are called as *gratification habits* and are considered by some as *masturbatory equivalents*. Also, these habit disorders tend to be commoner in individuals with mental retardation.

Treatment

The treatment is by behavior modification techniques and treatment of underlying psychopathology, if present.

ELECTIVE (SELECTIVE) MUTISM

Elective mutism is characterized by the presence of marked, emotionally determined, selectivity in speaking, despite the presence of language competence in at least some situations.

Typically, it is first manifested in early childhood and is seen more often in girls. It is estimated to be present in 3-8/10,000 children. It may be associated with temperamental traits involving social anxiety, withdrawal, sensitivity or resistance. The child is often mute in front of strangers or at school. Other socio-emotional disturbances may also be present.

Intramuscular and intravenous administration provide much more reliable blood levels. On an average, the oral liquid dose produces a peak level at 1½ hours, while the intramuscular dose peaks at 30 minutes.

The antipsychotics are highly lipophilic and highly protein-bound. They easily enter areas with good blood supply like brain, lung, kidneys and fetus, and accumulate there. They are not dialyzable.

The half-lives of almost all antipsychotics are long and theoretically a single dose administration per day is enough to produce sustained therapeutic blood levels. However in practice, divided doses are given initially to avoid side effects. Later, an attempt is made to give the whole dose or a major part of total daily dose at night.

Steady state plasma levels are usually reached in 5-10 days. Once the drug is withdrawn, it may remain in body for many days to many months.

The main metabolic pathway is through liver (hepatic microsomal enzymes). Oxidation and conjugation are the most important methods of metabolism for phenothiazines. Many of the metabolites, e.g. mesoridazine (for thioridazine), reduced haloperidol (for haloperidol) and 9-hydroxy-risperidone (for risperidone), are also active compounds. Chlorpromazine has more than 150 metabolites, some of which are active. The excretion of the metabolites is through kidneys and liver (entero-hepatic circulation).

Most of the antipsychotics tend to have a *therapeutic window*. If the blood level is below this 'window', the drug is ineffective. If the blood level is higher than the upper limit of the 'window', there is toxicity or the drug is again ineffective.

Classification

The currently available antipsychotic drugs can be classified as stated in Table 15.2.

Mechanism of Action

The exact mechanism of action of antipsychotics is unknown. But, most probably, one of the major mechanisms is the *antidopaminergic* activity of these drugs.

Antipsychotic drugs block D₂-receptors, the dopamine receptors which are mainly present in the mesolimbic system (mesolimbic system is concerned with emotional reactions) and the nigro-striatal system. The relative potencies of these drugs in competing for D₂ receptors parallel quite closely their clinical potencies.

It is currently believed that the antipsychotic drugs are effective in treating psychosis due to their action on the D₂ receptors located in the mesolimbic system, while the extrapyramidal side-effects (EPSE) are caused by the blockade of the D₂ receptors situated in the nigro-striatal system.

However, other neurotransmitters (e.g. 5-HT, acetylcholine) are clearly implicated (see below; atypical antipsychotics).

Sedation is caused by histaminergic blockade, which is maximum for drugs like chlorpromazine and thioridazine.

Second Generation Antipsychotics

A search for an antipsychotic drug, which acts only on the mesolimbic system but has no effect on the nigro-striatal system, has led to the development of a heterogeneous group of drugs, collectively known as *atypical* or *newer antipsychotics*. These are also known as *second generation anti-psychotics* (SGAs) or *serotonin-dopamine antagonists* or (SDAs), like risperidone, quetiapine and olanzapine.

By definition, these drugs are effective antipsychotics without producing undesirable extrapyramidal side-effects (i.e. *antipsychotics without neuroleptic effect*), and do not cause elevation of the serum prolactin levels.

These are characterized by a selective limbic dopamine blockade, D₄ receptor blockade, or a

Side Effects

The antipsychotics are safe drugs with a high therapeutic index and wide margin of safety in routine clinical dosages. In spite of this safety, a wide range of side-effects do occur with the use of antipsychotics even in the therapeutic doses.

The side-effects are listed in Table 15.4 with their mechanisms of causation and management.

Treatment Adherence

Many patients with schizophrenia can be uncooperative, with a poor treatment adherence. So, it may at times be difficult to ensure proper administration of antipsychotics on an outpatient basis.

Since discontinuation of antipsychotic medication often leads to relapse, long-acting preparations of antipsychotics are valuable in the treatment. They may be given in a depot form, either intramuscularly or administered orally.

There are several different preparations available in the international market but the following preparations are used most often:

1. Risperidone Consta (25-50 mg IM every 2 weeks)
2. Fluphenazine decanoate (25-50 mg IM every 2-4 weeks).
3. Haloperidol decanoate (25-250 mg IM every 4 weeks).
4. Flupentixol decanoate (20-400 mg IM every 2-4 weeks).
5. Zuclopenthixol decanoate (200-400 mg IM every 2-4 weeks)

Clinical Use

Some general principles regarding the routine clinical use of antipsychotics include:

1. Although atypical antipsychotics should usually be preferred, the choice of antipsychotic medication should be individualized in a particular patient.
2. Whenever possible, the lowest effective dose should be used.

3. Routinely, only one antipsychotic should be used at one time. Rational polypharmacy should be reserved only for judicious treatment after non-response to single antipsychotics.
4. High doses of antipsychotics should be avoided.
5. Whenever appropriate, psychosocial management should be combined with antipsychotic treatment.

ANTIDEPRESSANT DRUGS

Antidepressants are those psychotropic drugs which are used for the treatment of depressive disorders. These have also been called as mood-elevators and *thymoleptics*.

Isoniazid (INH) was found to have mood elevating properties in some patients suffering from tuberculosis in 1951. Iproniazid, a MAO Inhibitor and a derivative of INH, was later (1958) introduced for the treatment of depression.

The first tricyclic antidepressant *imipramine* was used in 1958 by Thomas Kuhn. It was different from phenothiazines by only a replacement of sulphur with an ethylene linkage. After this small structural difference (discovered by chance), imipramine was no longer effective as an antipsychotic but instead was quite beneficial in depressed patients.

Since 1958, the number of antidepressants has been gradually increasing.

Antidepressants have no euphoriant effect when administered to normal, non-depressed individuals.

Indications

Presently, the indications for the use of antidepressants include:

A. Depression

1. Depressive episode (also called as major depression, MDP depression, and endogenous depression)

TABLE 15.6. SIDE-EFFECTS OF ANTIDEPRESSANTS

Category and Side Effect	Probable Cause	Maximum with (For example)	Minimum with (For example)	Management
I. AUTONOMIC SIDE-EFFECTS				
1. Dry mouth	Muscarinic	Amitriptyline	Fluoxetine	See Table for 'Side Effects of Antipsychotics' (Table 15.3)
2. Constipation	Cholinergic	Amitriptyline	Fluoxetine	
3. Cycloplegia	Blockade	Amitriptyline	Fluoxetine	
4. Mydriasis	- do -	Amitriptyline	Fluoxetine	
5. Urinary retention	- do -	Amitriptyline	Fluoxetine	
6. Delirium	- do -	Amitriptyline	Fluoxetine	
7. Aggravation of glaucoma (Narrow angle)	- do -	Amitriptyline	Fluoxetine	
8. Orthostatic hypotension	α_1 Adrenergic Blockade	Amitriptyline	Fluoxetine	See Table for 'Side Effects of Antipsychotics' (Table 15.3)
II. SEXUAL SIDE-EFFECTS				
1. Impotence	α_1 Adrenergic Blockade	Amitriptyline	Mirtazapine Bupropion	Cyproheptadine given 1 hour before, can reverse anorgasmia and retarded ejaculation caused by SSRIs.
2. Impaired/retarded ejaculation	- do -	Amitriptyline Paroxetine	Mirtazapine Bupropion	
3. Anorgasmia	5-HT Blockade	SSRIs TCAs	Bupropion Mirtazapine	
4. Priapism	Not known	Trazodone	Not known	
III. CNS EFFECTS				
1. Sedation	α Adrenergic Blockade	Amitriptyline	Protriptyline Fluoxetine	This side-effect may be beneficial. Otherwise decrease dose. Change drug. Give single dose only at night. Decrease dose. Change drug.
2. Tremor and other EPS	Serotonergic	Amoxapine	Not known	
3. Jitteriness syndrome	Adrenergic; Serotonergic	Imipramine Fluoxetine	MAOI	This is very common in first week of therapy. Add diazepam for first 1-2 weeks. Tolerance occurs. Slow withdrawal of drug.
4. Withdrawal syndrome (mild)	Dependence	Imipramine Paroxetine	Protriptyline Fluoxetine	
5. Seizures	Decreased threshold	Bupropion Amitriptyline	Nomifensine Fluoxetine	Decrease dose. Change drug.
6. Aggravation of psychosis	Sympatho-mimetic	Not known	Amoxapine	Stop drug. Re-evaluate.
7. Precipitation of mania	Sympatho-mimetic	Tricyclics	Not Known	Stop drug. Use mood stabilizers.
IV. CARDIAC SIDE EFFECTS				
1. Tachycardia	Anticholinergic	Amitriptyline	Trazodone	Decrease dosage. Use safer drugs in elderly and those with past history or coexisting heart disease.
2. Quinidine-like action (decreased conduction time)	Cardiotoxic	Amitriptyline	Fluoxetine	
3. EKG changes	- do -	Amitriptyline	Fluoxetine	

Contd.

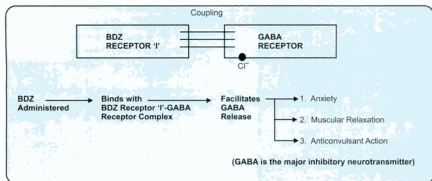


FIG. 15.1. PROBABLE MECHANISM OF ACTION OF BENZODIAZEPINES (BDZ)

Indications

The *indications* for use of carbamazepine include:

1. Seizures
 - i. Complex partial seizures (CPS)
 - ii. Generalized tonic clonic seizures
 - iii. Alcohol withdrawal seizures (rum fits), if persistent (also used sometimes for treatment of simple alcohol withdrawal syndrome)
2. Psychiatric disorders
 - i. Bipolar mood disorder (especially for rapid cyclers; lithium-refractory patients; lithium-intolerant patients; for the prophylaxis in some patients; for the treatment of acute mania, especially severe and dysphoric mania; for the treatment of acute depression, along with antidepressants or even alone)
 - ii. Impulse control disorder and aggression
 - iii. Psychosis (especially mania) with epilepsy
 - iv. Schizo-affective disorder
 - v. Borderline personality disorder
 - vi. Cocaine withdrawal syndrome
3. Paroxysmal pain syndromes
 - i. Trigeminal neuralgia
 - ii. Phantom limb pain
 - iii. Other paroxysmal pain syndromes (neuralgias).

Side Effects

The major side effects include diplopia, drowsiness, dizziness, nausea, vomiting, ataxia, skin rash, Steven-Johnson syndrome (erythema multiforme major), photosensitivity, cholestatic jaundice, acute oliguria with hypertension, leucopenia and thrombocytopenic purpura.

The most dangerous side-effects include bone marrow depression (causing aplastic anemia) and cardiovascular collapse.

So, it is essential to monitor cardiac, renal, hepatic and bone-marrow functions during carbamazepine treatment.

The use of carbamazepine in pregnancy and lactation should be best avoided.

ANTI-ANXIETY DRUGS

The anti-anxiety drugs, also known as the minor tranquilizers and anxiolytics, can be classified as follows:

not be used for more than 2 weeks at one time. Its safety in children, and in pregnancy and lactation is not proven.

5. Other Drugs

The other newer, hypno-sedative and anti-anxiety drugs include *suriclone* (a cyclopyrrolone

derivative; a hypnotic), *bretazenil* and *imidazenil* (partial benzodiazepine agonists; anxiolytic without sedation; rapid onset of action), *abecarnil* (β -carboline partial agonist at benzodiazepine receptor; anxiolytic and anticonvulsant), and *alpidem* (anxiolytic).

threshold (the minimum stimulus that induces an adequate seizure) and *calculation of the treatment stimulus* (usually 1.5 times supra-threshold for the bilateral and 2.5 times the unilateral ECT). So, the patient is stimulated at higher stimulus levels during the treatment than the seizure threshold (therefore supra-threshold).

After the seizure has occurred, mouth gag is removed, secretions are removed by a suction machine from the oral cavity, and oxygen mask is applied. Till consciousness is regained, the patient is turned to one side to prevent aspiration. The vital parameters are constantly monitored till recovery occurs. The patient is made to rest, for about 30 minutes to 1 hour, on bed after the treatment is over.

Duration of Therapy

The total duration and number of treatments given depends on the diagnosis, presence of side-effects, and the response to treatment.

Usually 6-10 treatments are sufficient, although up to 15 treatments can be given if needed. The treatments should be spaced, so that no more than 3 ECTs are given per week.

Although ECT is very effective in severe depressive disorder (for example), the benefit lasts only till the ECTs are given. There is no residual benefit after the treatment is over. Hence, the patient needs antidepressants (for example) during and after the ECTs are over.

Mechanism of Action

The efficacy of ECT is linked with the production of generalized tonic-clonic seizures though, as stated above, there is some recent doubt about the need for a seizure duration of 25-30 sec.

Although the exact mechanism is unclear, one hypothesis states that ECT possibly affects the catecholamine pathways between the diencephalon (from where seizure generalization occurs) and the limbic system (which may be

responsible for mood disorders), also involving the hypothalamus.

As ECT increases the threshold for further seizures, it may paradoxically act as an anti-convulsant. ECT also causes down-regulation of β_1 receptors in cortex and hippocampus.

Side Effects

1. Side-effects associated with general anesthesia: *Deaths* during ECT are usually due to the general anesthesia, succinylcholine (in patients with deficiency of pseudo-cholinesterase) and drug-interactions.

According to the APA task force report (2001), the approximate mortality rate is 1:10,000 patients (or 1:80,000 treatments), which is similar to any operative procedure under anesthesia.

2. *Memory disturbances* (both anterograde and retrograde) are very common. These are usually mild and recovery occurs within 1-6 months after treatment. Unilateral ECTs cause much less memory disturbance than bilateral ECTs.
3. *Confusion* may occur in the post-ictal period. Like memory disturbances, confusion is much

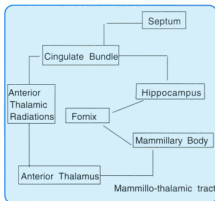


FIG. 16.1. PAPEZ CIRCUIT

TABLE 17.1. CONTD...

Defense Mechanism	Definition	Example in Normals	Illustration(s) from Clinical Situations
13. Hypochondriasis	Unconscious transformation of unacceptable impulses into inappropriate somatic concern	1. Abnormal illness behavior in physically disordered or normal individuals	1. Hypochondriasis
14. Inhibition	Involuntary decrease or loss of motivation to engage in some goal directed activity to prevent anxiety arising out of conflicts with unacceptable impulses	1. Writing 'blocks' or work 'blocks' 2. Social shyness	1. Obsessive compulsive disorder 2. Phobias
15. Compensation (Counterphobic defense)	Unconscious tendency to deal with a fear or conflict by unusual degree of effort in the opposite direction	1. Involvement in dare-devil activities (e.g. sky diving to counter fear of heights) 2. Excessive pre-occupation with body building to counter feelings of inferiority	1. Nymphomania (to counter a sense of sexual inadequacy) 2. Keeping excessive details in a diary in patients suffering from dementia
16. Splitting	Unconscious viewing of self or others as either good or bad without considering the whole range of qualities	1. Seeing all people without a mustache as feminine 2. Believing personalities to be either 'black' or 'white' without the shades of 'grey' (e.g. in a 'typical' Bollywood movie, the Hero is all good and the Villain all bad)	Borderline personality disorder
D. MATURE			
1. Sublimation	Unconscious gradual channelization of unacceptable infantile impulses into personally satisfying and socially valuable behavior patterns	1. Channelization of sexual or aggressive impulses into creative activities (e.g. diverting forbidden sexual impulses into artistic paintings)	—
2. Suppression (Voluntary)	Voluntary postponement of focusing of attention on an impulse which has reached conscious awareness	1. Voluntary decision not to think about an argument with parents while going for an interview	—
3. Anticipation	Realistic thinking and planning about future unpleasurable events	Anticipation is a universal phenomenon occurring in all intelligent persons	—
4. Humor	Overt expression of unacceptable impulses using humor in a manner which does not produce unpleasantness in self or others	A universal phenomenon	—
* No ego defense mechanism is psychotic, neurotic, immature, mature or normal per se. Almost all mechanisms of defense are sometimes used in normal individuals. It is an exclusive or abnormally excessive use that makes a defense mechanism neurotic or psychotic.			

The SD consists of three main steps:

- i. Relaxation training (described later).
- ii. Hierarchy construction.

Here the patient is asked to list all the conditions which provoke anxiety. Then, he is asked to list them in a descending order of anxiety provocation. Thus, a hierarchy of anxiety producing stimuli is prepared.

- iii. Systematic desensitization proper.

This can be done either in Imagery (SD-I) or in reality / in-vivo (SD-R).

At first, the lowest item in hierarchy is confronted (in reality or in imagery). The patient is advised to signal whenever anxiety occurs. With each signal, he is asked to relax (Step-I). After a few trials, patient is able to control his anxiety.

Thus, gradually the hierarchy is climbed till the maximum anxiety provoking stimulus can be faced in the absence of anxiety.

SD is a treatment of choice in phobias and obsessive-compulsive disorders.

B. Aversion Therapy

Aversion therapy is used for the treatment of conditions which are pleasant but undesirable, e.g. alcohol dependence, transvestism, ego dystonic homosexuality, other sexual deviations.

The underlying principle is the pairing of the pleasant stimulus (e.g. alcohol) with an unpleasant response (e.g. brief electrical stimulus), so that even in absence of the unpleasant response (after the therapy is over), the pleasant stimulus becomes unpleasant by association.

The unpleasant aversion is produced by electric stimulus (low voltage), drugs (e.g. apomorphine and disulfiram) or even by fantasy (when it is called as *covert sensitization*).

Typically, 20-40 sessions are given, with each session lasting about 1 hour. After completion of treatment, booster sessions may be given.

The current use of aversion therapy has declined sharply in the Western world as it is felt that it violates the human rights of the patient.

C. Operant Conditioning Procedures for Increasing a Behavior

The common methods for augmenting an adaptive behavior include:

- i. *Positive reinforcement.* Here, the desirable behavior is reinforced by a reward, either material or symbolic.
- ii. *Negative reinforcement.* Here, on performance of the desirable behavior, punishment can be avoided.
- iii. *Modeling.* The person is exposed to a 'model' behavior and is induced to copy it. This can also be used to avoid certain behaviors.

D. Flooding

This is a usually the method used in the treatment of phobias. Here, the person is directly exposed to the phobic stimulus, but escape is made impossible. By prolonged contact with the phobic stimulus, therapist's guidance and encouragement and therapist's modeling behavior, anxiety decreases and the phobic behavior diminishes.

E. Operant Conditioning Procedures for Decreasing a Behavior

These methods include:

- i. *Time-out.* Here, the reinforcement is withdrawn for some time, contingent upon the undesired response. Time-out is often used in therapy with children.
- ii. *Punishment.* Aversive stimulus is here presented, contingent upon the undesired response (i.e., whenever the undesired response occurs, punishment is given).
- iii. *Satiation.* The undesired response is positively reinforced, so that tiring occurs. A similar technique is *negative practice procedure*.

F. Other Behavioral Techniques

Last two decades have seen an explosion in the behavior therapy methodology. Many other

An *emergency* is defined as an unforeseen combination of circumstances which calls for an immediate action.

A *medical emergency* is defined as a medical condition which endangers life and/or causes great suffering to the individual.

A *psychiatric emergency* is a disturbance in thought, mood and/or action which causes sudden distress to the individual (or to significant others) and/or sudden disability, thus requiring immediate management.

A similar term *crisis* means a situation that presents a challenge to the patient, the family and/or the community.

TYPES OF PSYCHIATRIC EMERGENCIES

A psychiatric emergency can be one or more of the following:

1. A new psychiatric disorder with an acute onset.
2. A chronic psychiatric disorder with a relapse.
3. An organic psychiatric disorder.
4. An abnormal response to a stressful situation.
5. Iatrogenic emergencies.
 - i. Side-effects or toxicity of the psychotropic medication(s).
 - ii. Psychiatric symptomatology as a side-effect or toxicity of other medication(s).
6. Alcohol or drug dependence.
 - i. Withdrawal syndrome.
 - ii. Intoxication or Overdose.
 - iii. Complications.
7. Deliberate harm to self or others.

EXAMINATION

When faced with a psychiatric emergency, it is often important to combine speed with obtaining of 'complete' or 'adequate' information. A scheme for the typical emergency psychiatric evaluation is presented here:

I. Psychiatric History

It is important to 'always' obtain history from both the patient and the informant(s). Informant(s) may be more coherent and may provide more relevant information in emergency situation(s).

- i. *Chief complaint*: elaborate, with emphasis on dating of onset and progression.
- ii. *Recent life-changes*: like any losses (real or imaginary); any physical illnesses.
- iii. *Level of adjustment*, prior to the psychiatric emergency.
- iv. *Past history* (briefly) of any physical or psychiatric disorder(s).
- v. *Family history* (briefly) of any physical or psychiatric disorder(s).
- vi. *Drug and alcohol history*: prescription drugs, street drug(s) or alcohol dependence/abuse.

II. Detailed General Physical and Neurological Examination

It is essential to rule out (or diagnose) secondary psychiatric disorders, with particular emphasis on the presence of any head injury.

TABLE 19.5. ORGANIC AND PSYCHOGENIC STUPOR

	Organic Stupor	Psychogenic Stupor
1. Previous psychiatric illness	Usually absent	Usually present
2. Previous medical illness	Usually present	Usually absent
3. Previous history of stupor	10%	24-33%
4. Precipitating stressor	Usually absent	Usually present
5. Frequency	Much more common	Not so common (3-5% of all stupor cases)
6. Onset	Sudden or insidious	Often sudden
7. Course	Longer	Shorter. Recovery usually within 7 days
8. Protective reflexes (Blepharospasm, menace reflex, protective response)	Absent	Present
9. Resistance to eye opening	Absent	Usually present
10. Doll's head eye phenomenon (Oculocephalic reflex)	Usually present	Absent (avoids the gaze of examiner)
11. Meaningful posture and/or facial expressions	Absent	May be present
12. Urinary/fecal incontinence	Usually present	Unlikely
13. Neurological signs and deficits	May be present	Absent
14. Abnormal EEG	More often	Less often
15. Oculo-vestibular reflex	Absent	Present
16. Pentothal/amytal interview	Low dose (100-150 mg) increases stupor and neurological signs develop or increase	High dose (300-400 mg) increases alertness and a mental status examination can often be conducted
17. Mortality	35%	0-3%

pathology, then appropriate treatment can be instituted.

Examination

The examination consists of the following steps:

1. History and physical examination, with special emphasis on neurological examination.
2. Level of consciousness should ideally be rated on the Glasgow Coma Scale. This scale consists of three main categories of activity, namely eye opening (1-4), motor response

(1-5) and verbal response (1-5). The rating is done on a scale of 3 to 14.

3. Pentothal/amytal/diazepam interview is sometimes very helpful in differentiating organic and psychogenic catatonic stupor.
4. Investigations: Blood glucose, blood urea, serum creatinine, serum electrolytes, blood (arterial) gas analysis, EKG, blood and urine examination for drugs and poisons, blood and urine culture, and peripheral smear for malarial parasite.

the salient points regarding testamentary disposition are as follows:

1. A will must be in writing, though it need not be registered.
2. It must be signed by *testator* in the presence of at least two witnesses.
3. A legatee can not attest a will.
4. An executor(s) is appointed under the will by the testator to carry out its terms after his death.
5. A will can be revoked or modified any time before the death of the testator.
6. A will comes into effect after the death of the testator. It is said to speak from grave and to be 'ambulatory'.
7. The testator must be of a 'sound and disposing mind'. Section 59 of the Act states that 'every person of sound mind, not being a minor, may dispose of his property by will'.

Explanation 4 of this section states that 'no person can make a will while he is in such a state of mind, whether arising from intoxication or from illnesses or from any other cause, that he does not know what he is doing'.

If a medical practitioner is called to examine a testator as to his fitness to make a valid will, the following points must be kept in mind:

1. Testamentary capacity consists of:
 - i. an understanding of the nature of the will,
 - ii. a knowledge of the property to be disposed of, and
 - iii. an ability to recognize those who may have justifiable claims on his property.
2. The testator should be tested on the above mentioned points by thorough questioning.
3. If the testator is seriously ill, he must be made to read out aloud the will in the presence of the doctor.

4. A will is invalid if it is executed under undue influence of any other person. If there is reason to suspect that such is the case, the testator should be questioned when he is alone.
5. A will is invalid under the following conditions (for example):
 - i. imbecility arising from advanced age or by excessive drinking.
 - ii. insane delusions making the testator incapable of rational views and judgement.
6. A will is valid under the following conditions (for example):
 - i. deaf, dumb or blind persons who are not thereby incapacitated for making a will and are able to know what they do by it.
 - ii. lucid intervals.
 - iii. if testator commits suicide immediately after making the will, in the absence of evidence of mental disorder.
 - iv. presence of delusions not affecting in any way the disposal of the property or the persons affected by the will.
7. A will may be declared invalid if the testator disposes his property in a way which he would not have done under normal conditions.

V. Transfer of Property

Under the Transfer of Property Act 1882 (Section 7), only persons competent to contract, are authorized to transfer property.

VI. Contract

Under the Indian Contract Act 1872 (Section 11), every person to be competent to contract must be a major and of sound mind.

A person is said to be of sound mind for the purposes of a contract, if at the time of making a contract he is capable of understanding it and

For research purposes, consent should be entirely voluntary. The patient can withdraw consent at any stage, without this affecting patient's interest.

2. *Hospitalization* should be for patient's welfare, keeping in view the legal and administrative constraints, and social appropriateness.
3. *Psychiatric treatments* should be started only on clinical consideration for patient welfare and should be humane and never punitive. No psychiatrist should refuse to treat in an emergency.
4. No *gifts* and *gratifications* should be accepted from patients under treatment.
5. Any *sexual advance* towards any patient is unethical.
6. In case of doubt and/or unconventional treatment procedures a *second opinion* must be obtained.
7. It is unethical to force a *contract* on a patient during treatment.
8. Even if the patient is referred by legal or administrative authority, *patient's welfare* is paramount and patient should be informed of the purpose for which he is to be examined.

9. The basic *human rights* of mentally retarded should not be unethically abridged.

10. In the interest of the patient and the society, drug abusers who refuse to give consent may be treated with the consent of the their relatives. Effort has to be made to motivate them for accepting treatment voluntarily.

ICMR GUIDELINES FOR RESEARCH, 2000

In 2000, the Indian Council of Medical Research (ICMR) has released '*Ethical guidelines for biomedical research in human subjects*'.

These guidelines are very similar to the GCP (Good Clinical Practice) guidelines and the prevalent international guidelines. These guidelines regulate all biomedical research in human subjects in India.

MCI CODE OF MEDICAL ETHICS, 2002

All psychiatrists as doctors are also bound by the Code of medical ethics of Medical Council of India (MCI). Called as the *Indian Medical Council (Professional conduct, Etiquette and Ethics) Regulations, 2002*, the code came in to effect from 6th April, 2002. The full code is available at the website of MCI, at <http://mohfw.nic.in/code.htm>.

Stereotypy: Odd, repetitive and non-goal directed movement (can also be verbal).

Stupor: A clinical syndrome of akinesia and mutism, with relative preservation of conscious awareness.

Suicide: A human act of self-intentioned and self-inflicted cessation (death).

Tangentiality: Sudden and oblique digression in to unnecessary details that completely distract from the central theme; however, the patient never returns back to the original theme after digression (unlike in circumstantiality).

Thinking: Normal thinking is a goal directed flow of ideas, symbols and associations initiated by a problem or a task, characterized by rational connections between successive ideas or thoughts, and leading towards a reality oriented conclusion.

Thought Diffusion or Broadcasting: Subject experiences that his thoughts are escaping the confines of his self and are being experienced by others around.

Thought Echo: Voices speaking out thoughts aloud; also called as *echo de la pensee*.

Thought Insertion: Subject experiences thoughts imposed by some external force on his passive mind.

Thought Withdrawal: Thoughts cease and subject experiences them as removed by external force.

Tic: Involuntary, sudden, rapid, recurrent, non-rhythmic and stereotyped movement (can also be verbal).

Trance: A sleep like state of reduced consciousness and suggestibility.

Verbigeration: Senseless repetition of same words or phrases over and over again.

Waxy Flexibility: Parts of body can be placed in positions that will be maintained for long periods of time, even if very uncomfortable; flexible like wax. The process is called as *waxy flexibility*, while the end result is called as *cataplexy*.

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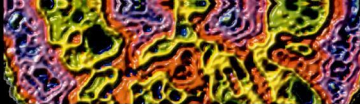
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A Short Textbook of **PSYCHIATRY**

A Short Textbook of Psychiatry aims to provide a brief yet comprehensive account of the psychiatric disorders and their allied aspects in a 'user-friendly' and 'easy-to-follow' manner.

In the span of last sixteen and a half years, the Short Textbook of Psychiatry has enjoyed a wide distribution among the undergraduate medical students, interns, junior residents, postgraduate psychiatry students, nursing students, psychology and psychiatric social work students, occupational therapy and physiotherapy students, general medical practitioners, other physicians and health professionals in India and abroad.

The sixth edition of the Short Textbook of Psychiatry has been once again extensively revised and updated.

Significant changes have been made in almost all the chapters, especially in chapters on Diagnosis and Classification (incorporating ICD-10 at most places and DSM-IV-TR wherever appropriate), Psychoactive Substance Use Disorders, Psychopharmacology, Schizophrenia, and Other Biological Methods of Treatment.

Colored Shaded boxes have been added at various places in the text to highlight the important points in tables and figures. The chapter on Psychiatric History and Examination contains a resume of laboratory tests in psychiatry, in additions to other significant changes. The Appendices have been revised and contain a glossary of common psychiatric terms.

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